

TICVAI Maintenance and Safety Management

Functional Requirements and Dashboard Design Specification

Eight boards covering asset management, preventive and corrective maintenance, work orders, safety, spare parts, resources, and analytics

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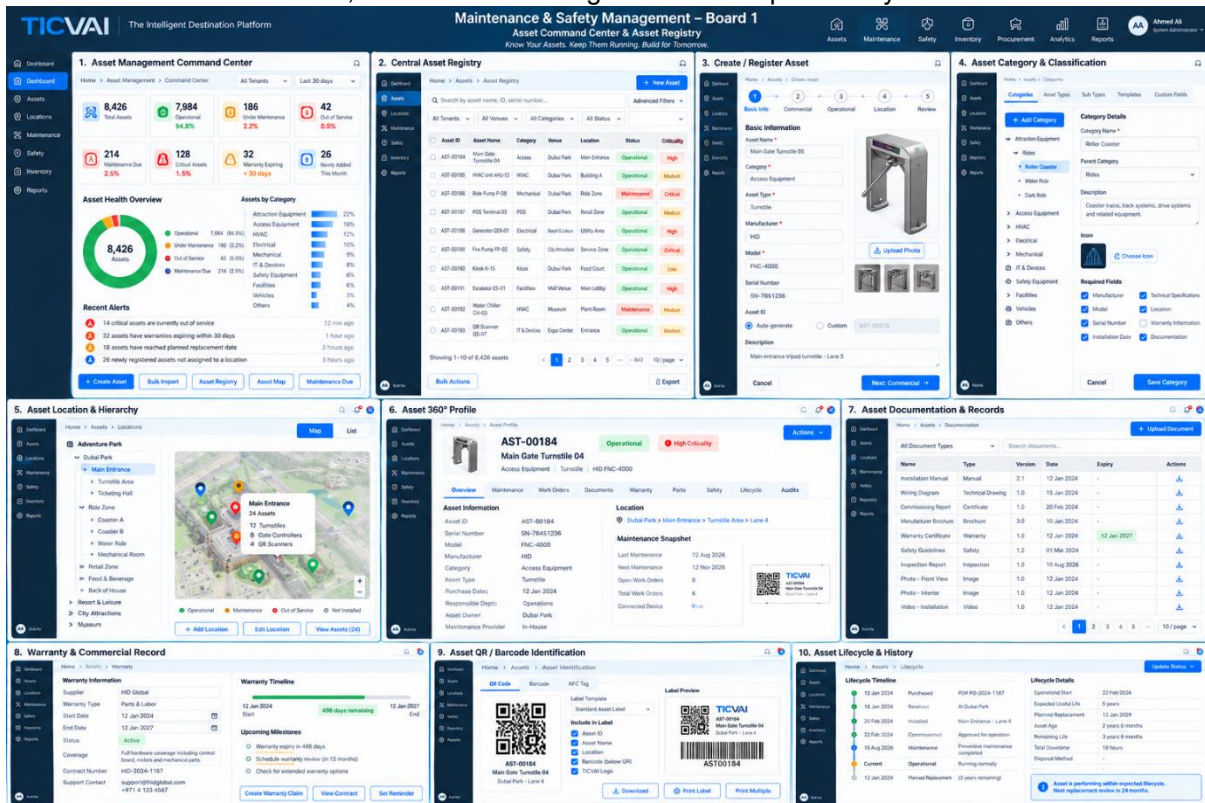
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Board 1 Asset Command Center and Asset Registry

Board Objective: Provide TICVAI with the centralized master repository and operational command center for managing all maintainable physical assets across tenants, venues, facilities, attractions, operational areas, and locations throughout their complete lifecycle.



Board 1 primarily covers **17.1 Asset Management**. The matrix requires the system to maintain a centralized asset registry and support asset categories, asset locations, lifecycle tracking, warranty tracking, documentation, and QR/barcode identification.

The key principle is:

Every maintainable physical asset must have one unique TICVAI asset record that becomes the single source of truth for its identity, location, ownership, technical information, documentation, warranty, maintenance history, and lifecycle.

This board is deliberately broader than Device Management. A connected device can be an asset, but **not every asset is a connected device**.

Screen 1 Asset Management Command Center

Purpose

Provide management, engineering, facilities, operations, and maintenance teams with a centralized overview of the entire physical asset estate.

KPI Cards

Display:

- Total Assets
- Operational Assets
- Out of Service
- Under Maintenance
- Maintenance Due
- Critical Assets
- Warranty Expiring
- End-of-Life Assets

- Newly Added Assets
- Unassigned Assets

Example:

KPI	Value
Total Assets	8,426
Operational	7,984
Under Maintenance	186
Out of Service	42
Maintenance Due	214
Critical Assets	128

Asset Health Overview

Provide breakdown by:

Tenant | Venue | Asset Category | Location | Operational Status | Criticality

Asset Categories

Examples:

- Attraction Equipment
- Ride Equipment
- Turnstiles
- Gates
- HVAC
- Electrical
- Mechanical
- Pumps
- Escalators/Lifts
- Generators
- POS/Kiosk Equipment
- IT/Connected Devices
- Safety Equipment
- Vehicles
- Facility Equipment
- Other

All categories must be configurable.

Attention Panel

Examples:

- 14 critical assets are currently out of service.
- 32 assets have warranties expiring within 30 days.
- △ 18 assets have reached their planned replacement date.
- 26 newly registered assets have not yet been assigned to a location.

Quick Actions

+ Create Asset

Bulk Import**Asset Registry****Asset Map****Warranty Center****Maintenance Due****Screen 2 Central Asset Registry****Purpose**

Provide the searchable master directory of all maintainable assets.

Asset Table

Asset ID	Asset Name	Category	Venue	Location	Criticality	Status
AST-00184	Main Gate Turnstile 04	Access	Dubai Park	Main Entrance	High	Operational
AST-00185	HVAC Unit AHU-12	HVAC	Dubai Park	Zone B	Medium	Operational
AST-00186	Ride Pump P-08	Mechanical	Dubai Park	Ride Zone	Critical	Maintenance

Search

Search using:

- Asset ID
- asset name
- serial number
- barcode
- QR code
- manufacturer
- model
- supplier reference

Filters

Support:

- Tenant
- Venue
- Building
- Area
- Zone
- Location
- Category
- Asset type
- Manufacturer
- Criticality
- Lifecycle status
- Operational status
- Warranty status
- Ownership

Bulk Actions

Authorized users can:

Assign

Move

Update Classification

Apply Maintenance Plan

Export

Print QR and Barcode

Screen 3 Create and Register Asset

Purpose

Provide a governed workflow for creating new asset records.

Basic Information

Capture:

- Asset name
- asset ID
- category
- asset type
- description
- manufacturer
- model
- serial number
- manufacturer asset number
- internal reference

Commercial Information

Where applicable:

- Supplier
- purchase date
- purchase order reference
- acquisition cost
- installation date
- commissioning date
- contract reference

Operational Information

- Criticality
- operational role
- responsible department
- asset owner
- maintenance owner
- maintenance provider

Location

Assign:

Tenant → Venue → Building/Area → Zone → Location

Registration Methods

Support:

Manual Registration**Bulk CSV and Excel Import****API Import****Device Manager Conversion and Link****Duplicate Detection**

Before creation, TICVAI checks:

- Serial number
- asset number
- device ID
- manufacturer/model
- existing QR/barcode

and warns about potential duplicate records.

Screen 4 Asset Category and Classification Management**Purpose**

Allow TICVAI administrators to define reusable asset structures instead of hardcoding every possible physical asset.

Category Hierarchy

Support:

Asset Category

↓

Asset Type

↓

Asset Subtype

Example:

Mechanical Equipment

→ **Pump**

→ **Water Circulation Pump**

or:

Access Equipment

→ **Turnstile**

→ **Tripod Turnstile**

Category Configuration

For each category configure:

- Required metadata
- technical fields
- maintenance attributes
- warranty fields
- safety classification
- criticality options
- documentation requirements

Dynamic Fields

Example:

HVAC Asset

May require:

- Cooling capacity
- refrigerant
- airflow
- power rating

Pump

May require:

- Flow rate
- pressure
- motor rating

Turnstile

May require:

- Gate type
- lane width
- controller
- direction

The form dynamically changes according to the asset type.

Templates

Administrators can create:

Asset Templates

Example:

Standard HVAC Unit Template

for rapid registration of similar assets.

Screen 5 Asset Location and Hierarchy

Purpose

Maintain the exact physical location and parent-child relationship of every asset.

Location Hierarchy

Support:

Tenant

↓

Venue

↓

Building and Attraction

↓

Area

↓

Zone

↓

Location

↓

Position

Example:

Adventure Park

→ Dubai Park

→ Roller Coaster A

- Mechanical Room
- Pump Zone
- Pump P-08

Parent and Child Asset Hierarchy

This is important for complex facilities.

Example:

Roller Coaster A

↓

Drive System

↓

Motor 01

↓

Gearbox 01

↓

Sensor 01

Each component can have its own asset record and maintenance history.

Asset Tree

Provide graphical hierarchy:

Parent Asset

- ├— Component 1
- ├— Component 2
- ├— Component 3
- └— Sub-component

Movement

Authorized users can:

Move Asset

with:

- Previous location
- new location
- date/time
- reason
- moved by
- approval where required

All movements must be preserved historically.

Screen 6 Asset 360 Profile

Purpose

Provide the single authoritative operational record for an asset.

Header

AST 00186 Ride Pump P 08

UNDER MAINTENANCE

Criticality: CRITICAL

Overview

Display:

- Asset ID
- asset name

- category
- manufacturer
- model
- serial number
- status
- owner

Location

Dubai Park → Ride Zone → Mechanical Room 02

Technical Information

Dynamic based on asset type.

Related Tabs

Overview

Maintenance

Work Orders

Documents

Warranty

Parts

Safety

Lifecycle

Costs

Audit History

Maintenance Snapshot

Display:

Last Maintenance: 10 Aug 2026

Next Maintenance: 10 Nov 2026

Open Work Orders: 1

Historical Failures: 4

Connected Device

Where applicable:

Linked Device: DEV-00186

Clicking it opens the Device Manager record.

This prevents duplicate asset/device information.

Screen 7 Asset Documentation and Digital Records

Purpose

Maintain all documentation associated with an asset.

The matrix explicitly requires asset documentation management.

Document Types

Support:

- User manuals
- maintenance manuals
- technical drawings
- wiring diagrams
- installation documents
- commissioning records

- certificates
- inspection reports
- warranty documents
- invoices
- purchase documents
- safety documents
- photographs
- videos
- vendor documents

Document Record

Capture:

Document Name

Type

Version

Issue Date

Expiry Date

Uploaded By

Visibility

Version Control

Example:

Maintenance Manual

V1.0 → V1.1 → V2.0

Historical versions remain available according to permissions.

Asset Context

Technicians should be able to open relevant documentation directly from the asset or work order.

Example:

Technician scans asset QR → opens asset → opens wiring diagram.

DAM Integration

Where TICVAI's Digital Asset Management module is deployed, large/common documents and media can be stored there while the Asset Management module maintains the association.

Screen 8 Warranty and Asset Commercial Record

Purpose

Track asset warranty coverage and related commercial/service information.

Warranty tracking is explicitly required by 17.1.

Warranty Information

Capture:

- Supplier
- manufacturer
- warranty provider
- warranty start
- warranty expiry
- warranty type
- coverage
- exclusions

- contract/reference
- support contact

Warranty Status

- ☒ **Active**
- ☐ **Expiring Soon**
- ☒ **Expired**
- ☐ **Not Available**

Warranty Alerts

Configure:

90 Days

60 Days

30 Days

7 Days

before expiry.

Service Contract

Where applicable:

- Contract number
- vendor
- start/end date
- SLA
- service scope
- contract documents

Maintenance Check

When creating corrective maintenance:

Asset is under warranty.

TICVAI can warn the user before internal repair is performed.

Actions

Create Warranty Claim

Contact Vendor

Attach Work Order

View Contract

Screen 9 Asset QR and Barcode Identification

Purpose

Provide fast physical identification of assets in the field.

The matrix explicitly requires asset QR-code and barcode identification.

Generate Identifier

Support:

QR Code

Barcode

and where future requirements justify it:

NFC Asset Tag

QR Record

Example:

AST-00186

Ride Pump P 08**Print**

Support:

Single Label**Bulk Labels****Label Template****Scan Experience**

Technician scans QR using a mobile/handheld interface.

TICVAI opens:

Ride Pump P 08

UNDER MAINTENANCE

Then provides authorized actions:

View Asset**Report Fault****Create Work Order****View Maintenance****View Manual****View Parts****Security**

The QR should identify the asset, but sensitive information and maintenance actions require authenticated access.

The QR itself should **not contain sensitive asset data**.

Screen 10 Asset Lifecycle and History**Purpose**

Track the asset from acquisition through commissioning, operation, maintenance, movement, replacement, and retirement.

Asset lifecycle tracking is explicitly required by the matrix.

Lifecycle Statuses

Configurable:

Planned

→ **Ordered**

→ **Received**

→ **Installed**

→ **Commissioning**

→ **Operational**

→ **Under Maintenance**

→ **Out of Service**

→ **Decommissioned**

→ **Retired / Disposed**

Lifecycle Timeline

Example:

10 Jan 2025

Purchased

↓

18 Jan 2025

Received

↓

22 Jan 2025

Installed

↓

24 Jan 2025

Commissioned

↓

18 Aug 2026

Corrective Maintenance

↓

20 Aug 2026

Returned to Service

↓

Planned Replacement

January 2030

Lifecycle Information

Capture:

- Purchase date
- received date
- installation date
- commissioning date
- operational start
- expected useful life
- replacement date
- decommission date
- retirement date
- disposal method

Asset Age

Display:

Asset Age: 1 year 7 months**Expected Useful Life: 5 years****Remaining Life: 3 years 5 months****Replacement Warning**

Example:

⚠ Asset has reached 85% of its planned useful life.

AI Recommendation

Where enough history exists:

“Pump P-08 has experienced increasing corrective-maintenance frequency and is approaching its planned replacement period. Review replacement versus continued repair.”

The final decision remains with authorized users.

Board 1 Shared Configuration Requirements

Authorized administrators shall configure:

- Asset categories
- asset types/subtypes
- asset numbering rules
- asset templates
- dynamic metadata fields
- manufacturer/model catalogue
- asset criticality levels
- ownership models
- responsible departments
- maintenance ownership
- venue/location hierarchy
- parent/child asset rules
- lifecycle statuses
- useful-life rules
- replacement thresholds
- warranty types
- warranty alerts
- documentation types
- required documents by asset category
- QR/barcode formats
- label templates
- movement reason codes
- retirement/disposal reasons
- approval requirements
- asset visibility rules

Criticality Classification

I recommend TICVAI include a configurable **Asset Criticality Model**.

Example:

Critical

Failure may:

- Stop venue/attraction operations
- create safety exposure
- significantly affect guests
- affect revenue
- create regulatory risk

High

Significant operational impact but redundancy may exist.

Medium

Limited operational impact.

Low

Minimal immediate operational impact.

Criticality should influence later boards:

Maintenance Priority**SLA****Escalation****Inspection Frequency****Spare Parts Strategy****Approval**

This means a failed decorative display should not be treated the same way as a failed ride safety component.

Asset vs Device Important Architecture

Softlabs must understand this distinction clearly.

Asset

A physical item maintained by the organization.

Examples:

Pump**HVAC Unit****Turnstile****Generator****Ride Motor****Fire Extinguisher****Escalator****Device**

A connected technology endpoint managed by Device Management.

Examples:

QR Reader**Gate Controller****POS Terminal****Game Reader****Kiosk**

Some objects can be **both**.

For example:

Turnstile Controller

Asset Record:

AST-00184

Device Record:

DEV-00422

These records should be **linked**, not duplicated.

The Asset record owns:

Maintenance / Warranty / Physical Lifecycle / Commercial Information

The Device record owns:

Connectivity / Configuration / Firmware / Security / Device Integration

This architecture is important for avoiding duplicate data across Modules 16 and 17.

Asset Hierarchy Architecture

I recommend supporting both:

Location Hierarchy

Tenant → Venue → Building → Area → Zone → Position

and:

Asset Hierarchy

System → Asset → Component → Sub-component

Example:

Roller Coaster

↓

Drive System

↓

Motor

↓

Gearbox

↓

Sensor

This becomes extremely important in later maintenance boards because a work order may affect only a component rather than the entire attraction.

RBAC and Governance

Permissions should distinguish:

View Assets

Create Asset

Edit Asset

Change Classification

Move Asset

Manage Documentation

Manage Warranty

Generate QR

Link Device

Change Lifecycle Status

Decommission Asset

Retire Asset

View Commercial Information

Critical lifecycle actions such as:

Decommission

Retire

Dispose

can require approval.

All changes must maintain:

Who | What | Before | After | When | Reason

Board 1 Acceptance Criteria

Board 1 is accepted when TICVAI can successfully:

Register → Classify → Locate → Identify → Document → Track Warranty → Link Components/Devices → Track Lifecycle → Decommission/Retire

all maintainable physical assets.

The system must provide a centralized asset registry and support asset categories, locations, lifecycle tracking, warranty tracking, documentation, and QR/barcode identification as required by the matrix.

It must also support multi-tenant/venue isolation, dynamic asset types, parent-child hierarchies, bulk registration, complete historical tracking, and governed lifecycle actions.

Board Boundary

To prevent duplication across the module:

Board 1 — Asset identity, registry, hierarchy & lifecycle

Board 2 — Preventive maintenance planning

Board 3 — Corrective & emergency maintenance

Board 4 — Work order execution

Board 5 — Safety & compliance

Board 6 — Spare parts & inventory

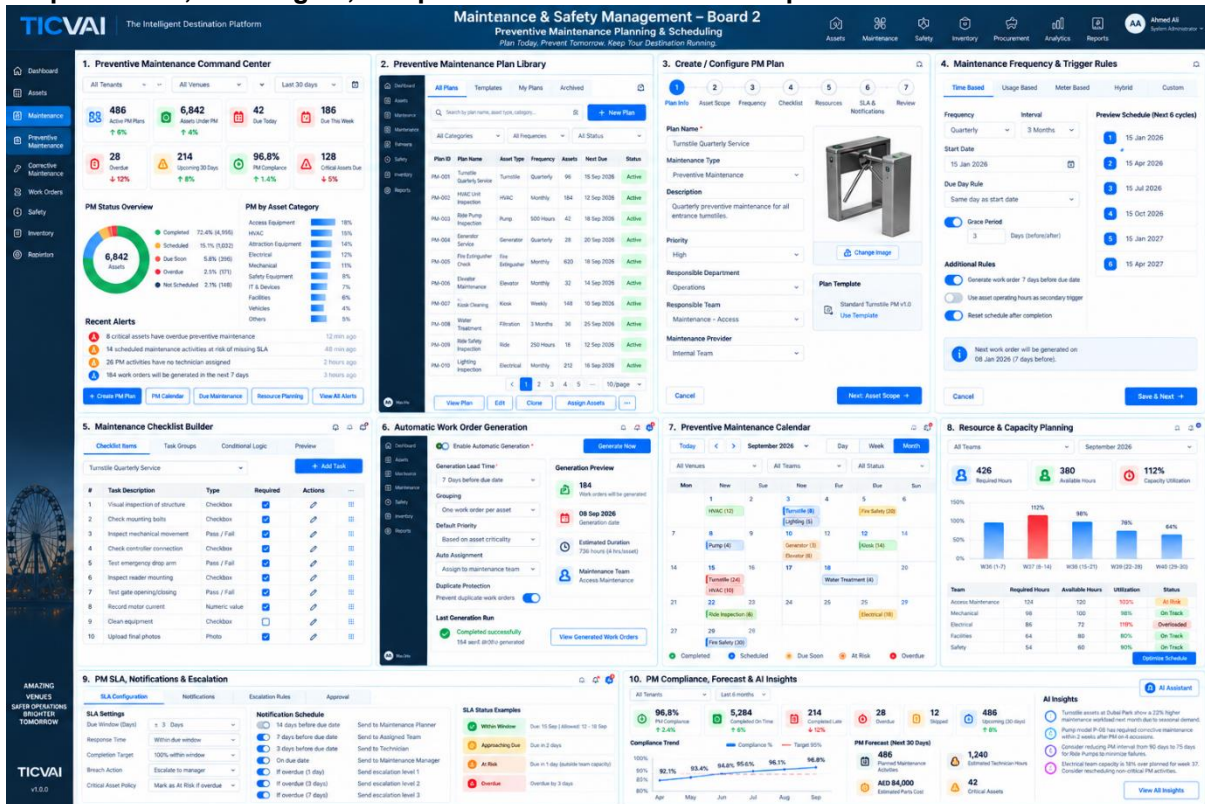
Board 7 — Procurement & maintenance resources

Board 8 — Intelligence, analytics & reporting

Board 1 = 10 screens.

Board 2 Preventive Maintenance Planning and Scheduling

Board Objective: Provide TICVAI with a centralized preventive-maintenance engine that allows maintenance teams to define maintenance plans, recurring schedules, maintenance frequencies, asset-specific checklists, automatic work-order generation, maintenance calendars, resource requirements, SLA targets, and preventive-maintenance compliance before asset failures occur.



Board 2 primarily covers **17.2 Preventive Maintenance**. The matrix requires preventive-maintenance plans and schedules, configurable maintenance frequencies, automatic generation of work orders, maintenance calendars, resource planning, SLA tracking, and maintenance notifications.

The core principle is:

TICVAI should move maintenance from “repair after failure” to “maintain before failure.”

Board 1 tells us **what the asset is**. Board 2 determines **what preventive maintenance it requires, when it must happen, and what resources are needed**.

Screen 1 Preventive Maintenance Command Center

Purpose

Provide maintenance managers with a centralized overview of all preventive-maintenance activity across tenants, venues, assets and locations.

KPI Cards

Display:

- Active PM Plans
- Assets Under PM
- Due Today
- Due This Week
- Overdue
- Upcoming 30 Days
- PM Work Orders
- PM Compliance %

- SLA at Risk
- Critical Assets Due

Example:

KPI	Value
Active PM Plans	486
Assets Covered	6,842
Due Today	42
Due This Week	186
Overdue	28
PM Compliance	96.8%

PM Status

- ☒ Completed
- ☐ Scheduled
- ☐ Due Soon
- ☒ At Risk
- ☒ Overdue
- ☐ Not Scheduled

Maintenance Overview

Break down by:

Tenant | Venue | Asset Category | Asset Criticality | Maintenance Type | Team

Attention Panel

Examples:

- ☒ 8 critical assets have overdue preventive maintenance.
- ☒ 14 scheduled maintenance activities are at risk of missing SLA.
- ☐ 26 PM activities have no technician assigned.
- ☐ 184 work orders will be generated during the next seven days.

Quick Actions

+ Create PM Plan

PM Calendar

Due Maintenance

Resource Planning

PM Compliance

Screen 2 Preventive Maintenance Plan Library

Purpose

Provide a centralized library of reusable preventive-maintenance plans.

PM Plan Directory

Plan	Asset Type	Frequency	Assets	Next Due	Status
Turnstile Quarterly Service	Turnstile	Quarterly	96	15 Sep	Active
HVAC Monthly Inspection	HVAC	Monthly	184	12 Sep	Active

Plan	Asset Type	Frequency	Assets	Next Due	Status
Ride Pump Inspection	Pump	500 hrs	42	Usage	Active
Fire Extinguisher Inspection	Safety	Monthly	620	18 Sep	Active

Search and Filters

Filter by:

- Tenant
- venue
- asset category
- asset type
- maintenance type
- frequency
- criticality
- status
- responsible team

PM Plan Status

Draft

→ Pending Approval

→ Active

→ Suspended

→ Archived

Actions

Create

Edit

Clone

Assign Assets

Suspend

Archive

View History

Reusable Templates

TICVAI should allow:

PM Template → **Apply to multiple assets**

rather than creating individual maintenance plans repeatedly.

Screen 3 Create and Configure Preventive Maintenance Plan

Purpose

Provide a structured wizard for creating preventive-maintenance requirements.

Step 1 Plan Information

Capture:

- Plan name
- maintenance type
- description
- priority
- criticality
- responsible department

- responsible team
- maintenance provider

Step 2 Asset Scope

Apply to:

Single Asset

Selected Assets

Asset Group

Asset Type

Asset Category

Location

Venue

Example:

All HID Turnstiles – Dubai Park

Step 3 Maintenance Frequency

Select maintenance trigger.

Step 4 Checklist

Assign maintenance tasks.

Step 5 Resources

Define:

- Technician skill
- team
- tools
- parts
- estimated duration

Step 6 SLA and Notification

Configure:

- due window
- SLA
- reminders
- escalation

Step 7 Review

Show:

Assets affected

Estimated annual work orders

Estimated maintenance hours

Next scheduled maintenance

Actions

Save Draft

Submit for Approval

Activate

depending on permissions.

Screen 4 Maintenance Frequency and Trigger Rules

Purpose

Allow TICVAI to support different preventive-maintenance strategies instead of relying only on fixed calendar dates.

Time Based Maintenance

Support:

Daily

Weekly

Monthly

Quarterly

Semi Annual

Annual

Custom Interval

Example:

Every 90 days.

Usage Based Maintenance

Where operational data exists:

Every 500 operating hours.

Every 100,000 cycles.

Every 250,000 scans.

Every 50,000 gate openings.

Meter Based Maintenance

Support meters such as:

- Runtime hours
- distance
- cycles
- transactions
- operating count
- manufacturer-specific meter

Fixed Date Maintenance

Example:

Every 1 January and 1 July.

Hybrid Trigger

This is important.

Support:

Every 3 months OR 50,000 cycles — whichever comes first.

Trigger Reset

After completed maintenance:

Reset Maintenance Counter

where applicable.

Grace Window

Example:

Due: 15 September

Allowed Window: ± 3 days

This gives operations flexibility without automatically classifying a job as overdue.

Screen 5 Preventive Maintenance Checklist Builder**Purpose**

Define exactly what technicians must inspect, service, test or replace during preventive maintenance.

Checklist Example**Turnstile Quarterly Maintenance**

- ☐ Visual inspection
- ☐ Check mounting bolts
- ☐ Inspect mechanical movement
- ☐ Check controller connection
- ☐ Test emergency drop arm
- ☐ Inspect reader mounting
- ☐ Test gate opening
- ☐ Test gate feedback
- ☐ Clean equipment
- ☐ Record final condition

Task Types

Support:

Checkbox**Pass and Fail****Numeric Reading****Text****Photo Required****Video****Measurement****Signature****Attachment****Measurement Example****Motor Temperature**

Expected:

$\leq 65^{\circ}\text{C}$

Technician enters:

72°C

TICVAI flags:

● **Outside Allowed Range**

Conditional Checklist

Example:

If "Oil Leakage = YES"

automatically show:

Upload Photo**Record Leakage Location****Create Corrective Maintenance Request****Mandatory Tasks**

Critical tasks cannot be skipped without:

Reason Authorization**Screen 6 Automatic Work Order Generation****Purpose**

Automatically convert scheduled preventive-maintenance activities into executable work orders.

Automatic generation of maintenance work orders is explicitly required by the matrix.

Generation Rules

Example:

PM Due Date: 15 September

Generate Work Order: 7 days before

Therefore:

WO Created: 8 September

Work Order Information

Automatically populate:

- Asset
- maintenance plan
- checklist
- venue/location
- priority
- estimated duration
- required skill
- required parts
- SLA
- target completion

Generation Options

One WO per Asset

or

Grouped WO

Example:

Inspect 20 fire extinguishers in Zone A.

Assignment

Automatically assign based on:

Team

Technician Skill

Venue

Shift

Availability

or leave:

Unassigned

for maintenance planning.

Duplicate Protection

TICVAI must prevent duplicate work orders being generated for the same PM occurrence.

Linkage

Maintain:

PM Plan → PM Schedule → Work Order → Execution → Completion

Board 4 will manage the detailed work-order execution.

Screen 7 Preventive Maintenance Calendar

Purpose

Provide maintenance teams with a visual operational calendar of upcoming maintenance activities.

Views**Day****Week****Month****Timeline****Venue****Calendar Cards**

Example:

15 SEP**Main Entrance****Turnstile Quarterly Service**

12 Assets

Team: Maintenance A**Estimated:** 4 Hours**Color and Status** Completed Scheduled Due Soon At Risk Overdue Suspended**Filters**

Filter by:

- Tenant
- venue
- building
- area
- asset
- category
- criticality
- team
- technician
- PM plan
- status

Drag and Reschedule

Authorized users may reschedule activities.

Before rescheduling, show:

Current Date: 15 Sep

Proposed Date: 18 Sep

SLA: Will remain within allowed window ✓

or:

⚠ Proposed date will create an SLA breach.

Bulk Rescheduling

Support planned events such as:

Venue Closure

Public Holiday

Major Event

Operational Blackout

Screen 8 Maintenance Resource and Capacity Planning

Purpose

Ensure that TICVAI does not create a maintenance schedule that cannot realistically be executed.

Resource Demand

Display:

Scheduled Maintenance Hours

versus

Available Technician Hours

Example:

Week 38

Required:

426 hours

Available:

380 hours

Capacity Warning

● Maintenance demand exceeds available capacity by 46 hours.

Resource Requirements

Each PM plan can require:

- Skill
- certification
- technician level
- team
- number of technicians
- tools
- equipment
- estimated duration
- parts

Resource Calendar

Show:

Team	Mon	Tue	Wed	Thu	Fri
Electrical	82%	94%	112%	76%	68%
Mechanical	64%	88%	92%	71%	54%

Capacity Actions

Reassign

Reschedule

Add Resource

Split Work

External Vendor

Detailed technician/resource operations will be expanded in Board 7, while this screen provides the planning context needed for PM scheduling.

Screen 9 PM SLA Notifications and Escalation

Purpose

Ensure preventive-maintenance tasks are completed within their required maintenance window.

PM SLA Example

Critical Ride Pump

PM Due: 15 Sep

Allowed Completion: 12–16 Sep

Current Status: Scheduled

Notification Rules

Example:

14 Days Before

→ Maintenance Planner

7 Days Before

→ Assigned Team

3 Days Before

→ Technician

Due Date

→ Maintenance Manager

Overdue

→ Escalate

SLA Status

☒ Within Window

☐ Approaching Due

☒ At Risk

☒ Overdue / Breached

Escalation

Example:

PM overdue 1 day

→ Maintenance Supervisor

3 days

→ Maintenance Manager

7 days

→ Venue Operations Manager

Critical Asset Escalation

For safety-critical assets:

Missed preventive maintenance may automatically change the asset to **Restricted / Inspection Required**, depending on configured policy.

TICVAI should not automatically shut down an asset unless a governed operational rule explicitly requires it.

Screen 10 PM Compliance Forecast and AI Recommendations

Purpose

Provide management with preventive-maintenance performance and forward-looking maintenance intelligence.

KPI Cards

Display:

- PM Compliance %
- Completed On Time
- Completed Late
- Overdue
- Skipped
- Rescheduled
- Upcoming 30 Days
- Estimated Maintenance Hours

Compliance Formula

Example:

Completed Within Required Window / Total Due PM Activities

Trend

Show:

Jan → Sep

Example:

93.4% → 94.8% → 96.8%

Compliance Analysis

Analyze by:

Tenant

Venue

Asset Category

Asset Type

Criticality

Team

Maintenance Plan

Maintenance Forecast

Example:

Next 30 Days

486 PM Activities

1 240 Technician Hours

AED 84 000 Estimated Parts

42 Critical Assets

AI Insights

Examples:

“Mechanical Team A is projected to exceed available maintenance capacity by 18% next week.”

“Pump model P-X8 has required corrective maintenance shortly before its scheduled quarterly PM on six occasions.”

“Moving P-X8 preventive maintenance from 90 days to approximately 75 days may reduce unplanned failures.”

“12 critical assets have PM dates overlapping with the venue's peak operating period.”

AI Actions

Review Frequency

Reschedule

Increase Resources

Review Asset Reliability

Create Proposed Plan Change

AI should recommend changes, but changing an approved maintenance frequency must follow governance/approval.

Board 2 Shared Configuration Requirements

Authorized administrators shall be able to configure:

- Preventive-maintenance types
- PM plan templates
- frequency types
- calendar frequencies
- usage-based triggers
- meter-based triggers
- hybrid triggers
- maintenance windows
- grace periods
- asset applicability rules
- maintenance checklists
- checklist field types
- acceptable measurement ranges
- conditional tasks
- mandatory tasks
- work-order generation lead time
- grouped vs individual work orders
- default priorities
- resource requirements
- technician skills
- estimated duration
- SLA rules
- notification rules
- escalation rules
- rescheduling permissions
- PM suspension rules
- approval workflows
- compliance calculation
- forecasting horizon

Preventive Maintenance Hierarchy

I recommend Softlabs structure PM configuration as:

PM Template

↓

Asset Category and Type

↓

Tenant

↓

Venue

↓

Asset Group

↓

Individual Asset Override

For example:

Global Turnstile PM Template**Quarterly Inspection**

↓

Adventure Park

Same template

↓

Dubai Park

Every 90 days

↓

Main Entrance TurnstilesEvery **60 days** because of heavier usage

This allows standardization without removing venue-level flexibility.

Critical Requirement Calendar vs Usage Based Maintenance

Softlabs should not design Board 2 as only a calendar scheduler.

For TICVAI's target venues, maintenance may depend heavily on **actual asset usage**.

For example:

Turnstile A

Opened:

1 000 000 times**Turnstile B**

Opened:

150 000 times

Both may have been installed on the same date.

Maintaining both only every three months may not make sense.

Therefore the PM engine should support:

Time

•

Usage

•

Condition and Meter

and:

Whichever Comes First

This will also allow future AI/predictive maintenance without redesigning the PM engine.

Relationship With Device Management

For connected assets, Board 2 can consume telemetry from Device Management.

Example:

Gate Controller GC 04

Device Manager reports:

Opening Cycle Count = 98,500

Preventive Maintenance rule:

Service at 100,000 cycles

TICVAI predicts:

Approximately 3 days until PM threshold.

Then:

PM Engine

↓

Automatically generates Work Order

There must still be:

One Asset One Maintenance Plan One Work Order

rather than a separate maintenance record inside Device Management.

Relationship With Boards 3 and 4

The distinction is important:

Board 2 Preventive

“We expect maintenance to be required.”

Board 3 Corrective and Emergency

“Something has failed or a problem has been detected.”

Board 4 Work Order

“Someone now needs to perform the work.”

Therefore:

Preventive Plan

↓

Schedule and Trigger

↓

Work Order

↓

Board 4 Execution

If the technician discovers a fault during preventive maintenance:

PM Work Order

↓

Defect Identified

↓

Board 3 Corrective Maintenance

↓

Corrective Work Order

This maintains full traceability.

RBAC and Governance

Permissions should distinguish:

View PM Plans

Create PM Plan

Edit PM Plan

Approve PM Plan

Activate PM Plan

Assign Assets

Change Frequency

Build Checklist

Schedule Maintenance

Reschedule

Assign Resources

Suspend PM

Override Due Date

View PM Compliance

Changes to maintenance frequency for **critical/safety assets** should optionally require additional approval.

All changes must preserve:

Before | After | User | Time | Reason | Approval

Board 2 Acceptance Criteria

Board 2 is accepted when TICVAI can successfully:

Define PM Plan → Assign Assets → Set Frequency/Trigger → Build Checklist → Plan Resources → Schedule → Generate Work Order → Notify/Escalate → Track Compliance → Forecast

preventive maintenance across the asset estate.

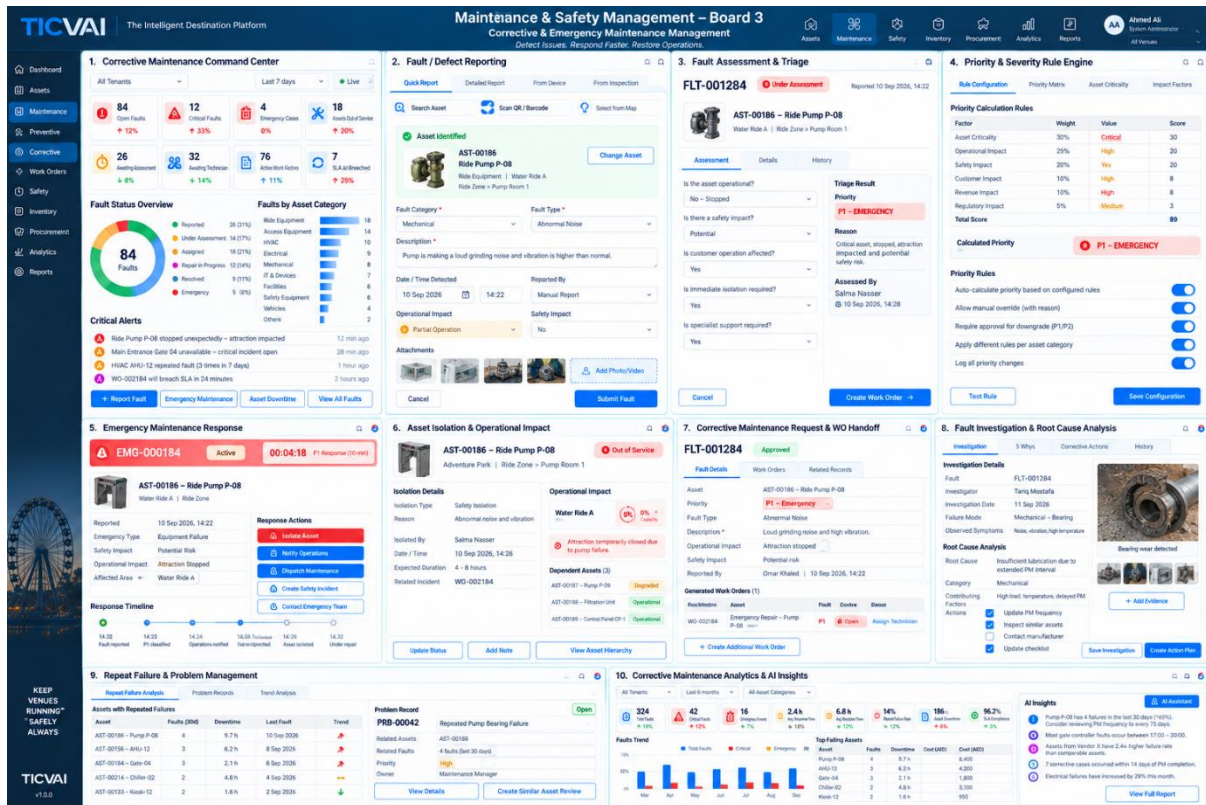
The system must support preventive-maintenance plans and schedules, configurable maintenance frequencies, automatic work-order generation, maintenance calendars, resource planning, SLA tracking and notifications as required by the matrix.

It must additionally support **time-based, usage-based, meter-based and hybrid maintenance triggers**, reusable templates, multi-tenant/venue policies, resource-capacity checks, configurable checklists, critical-asset governance and complete historical auditability.

Board 2 = 10 screens.

Board 3 Corrective and Emergency Maintenance Management

Board Objective: Provide TICVAI with a centralized operational system for identifying, reporting, assessing, prioritizing, escalating, and resolving **asset faults, breakdowns, defects, and emergency maintenance events.**



Board 3 covers the corrective side of maintenance and connects directly with Board 4 for detailed work-order execution.

The key principle is:

When something fails, TICVAI must immediately understand what failed, how serious it is, what operations are affected, who needs to respond, and how quickly action is required.

The separation between the boards should remain clear:

Board 2 = Prevent the failure

Board 3 = Detect, report, assess and manage the failure

Board 4 = Execute and close the maintenance work

Screen 1 Corrective Maintenance Command Center

Purpose

Provide maintenance and operations teams with a real-time overview of all active faults, breakdowns, defects, and emergency maintenance activities.

KPI Cards

Display:

- Open Faults
- Critical Faults
- Emergency Cases
- Assets Out of Service
- Awaiting Assessment
- Awaiting Technician
- Active Corrective Work Orders

- Repeat Failures
- SLA at Risk
- SLA Breached

Example:

KPI	Value
Open Faults	84
Critical	12
Emergency	4
Out of Service	18
SLA at Risk	9
Repeat Failures	7

Fault Status Overview

Show:

- ☐ Reported
- ☒ Under Assessment
- ☐ Assigned
- ☒ Repair in Progress
- ☐ Resolved
- ☒ Critical / Emergency

Breakdown

Analyze by:

Tenant | Venue | Asset Category | Location | Criticality | Fault Type | Priority

Critical Attention Panel

Examples:

- ☒ Ride Pump P-08 stopped unexpectedly — attraction impacted.
- ☒ Main Entrance Gate 04 unavailable — critical incident open.
- ☒ HVAC AHU-12 has generated the same fault three times in seven days.
- △ Corrective WO-002184 will breach SLA in 24 minutes.

Quick Actions

+ Report Fault

Emergency Maintenance

Open Faults

Asset Downtime

Repeat Failures

Screen 2 Fault and Defect Reporting

Purpose

Allow authorized users, technicians, operational teams, APIs, and connected systems to report an asset problem quickly.

Report Fault

Capture:

- Asset

- tenant
- venue
- exact location
- fault category
- fault type
- description
- date/time detected
- reported by
- detection source
- operational impact
- safety impact
- attachments

Reporting Methods

Support:

Web Portal

Mobile and Technician Interface

QR and Barcode Scan

Device Monitoring

API

Inspection

Preventive Maintenance

QR Flow

Technician scans:

AST-00186

↓

Ride Pump P 08

↓

Report Fault

This should automatically populate the asset and location.

Evidence

Allow:

Photo

Video

Document

Voice and Technician Note

Quick Report

For field operations:

Scan Asset → Select Fault → Add Photo → Submit

The user should not need to complete a long administrative form just to report a fault.

Screen 3 Fault Assessment and Triage

Purpose

Determine the seriousness of a reported problem before maintenance work begins.

Fault Assessment

Display:

Asset: Ride Pump P-08

Criticality: CRITICAL

Fault: Abnormal Pressure

Reported: 14:18

Assessment Questions

Configure questions such as:

Is the asset operational?

Yes / Limited / No

Is there a safety impact?

Yes / No / Unknown

Is customer operation affected?

Yes / No

Is immediate isolation required?

Yes / No

Is specialist support required?

Yes / No

Assessment Outcome

TICVAI determines:

Impact

Urgency

Priority

Example

Asset Criticality:

Critical

Operational Status:

Stopped

Safety Impact:

Potential

Result:

● P1 — EMERGENCY

Actions

Create Work Order

Escalate

Isolate Asset

Create Safety Incident

Request Specialist

Screen 4 Priority and Severity Rule Engine

Purpose

Automatically determine maintenance priority using standardized rules rather than relying only on subjective user selection.

Priority Levels

Recommended:

P1 Emergency

Immediate safety or major operational impact.

P2 Critical

Major operational impact requiring urgent action.

P3 High

Significant problem but limited temporary operation may continue.

P4 Normal

Standard corrective maintenance.

P5 Low

Minor defect with limited operational impact.

Priority Inputs

Calculate using:

Asset Criticality

-

Operational Impact

-

Safety Impact

-

Customer Impact

-

Revenue Impact

-

Regulatory Impact**Example Rule**

If:

Asset Criticality = Critical

AND

Asset Status = Stopped

AND

Safety Impact = Yes

Then:

Priority = P1 Emergency

Manual Override

Authorized users may override the calculated priority.

Require:

Reason

and optionally:

Approval

The original system-calculated priority must remain in the audit history.

Screen 5 Emergency Maintenance Response**Purpose**

Provide a dedicated high-priority workflow for emergencies requiring immediate operational action.

Emergency Header

● **EMERGENCY MAINTENANCE**

EMG-000184

Asset: Ride Pump P-08

Venue: Adventure Park

Location: Ride Zone

Emergency Information

Display:

- Emergency type
- time detected
- affected asset
- safety impact
- operational impact
- affected attraction/area
- assigned response team
- elapsed time

Immediate Actions

Depending on permission:

Isolate Asset

Stop Operation

Notify Operations

Dispatch Maintenance

Create Safety Incident

Contact Emergency Team

Create Emergency Work Order

Response Timer

Example:

P1 Response SLA

00:04:18 elapsed

Target:

10 minutes

Emergency Timeline

14:18 — Fault detected

14:19 — P1 classified

14:20 — Operations notified

14:21 — Technician dispatched

14:24 — Asset isolated

Important Governance

TICVAI should not automatically shut down physical equipment unless an approved integration and safety policy explicitly permits remote action.

Otherwise, TICVAI issues:

IMMEDIATE ISOLATION REQUIRED

and tracks confirmation.

Screen 6 Asset Isolation and Operational Impact

Purpose

Manage assets that cannot safely or operationally remain in normal service.

Asset Status

Allow:

☒ Operational

- Restricted Operation
- ◐ Degraded
- ◑ Out of Service
- Isolated / Locked Out

Isolation Record

Capture:

- Asset
- reason
- isolation type
- isolated by
- date/time
- expected duration
- safety requirement
- authorization
- related incident
- related work order

Operational Impact

Record affected:

- Attraction
- admission lane
- ride
- facility
- retail point
- F&B location
- building
- operational zone

Example

Ride Pump P 08

Status:

OUT OF SERVICE

Affected:

Water Ride A

Operational Capacity:

0

Dependency Awareness

If asset hierarchy exists:

Pump P 08

↓

belongs to:

Water System

↓

supports:

Water Ride A

TICVAI should show this impact chain.

Screen 7 Corrective Maintenance Request and Work Order Handoff

Purpose

Convert an assessed fault into an executable maintenance work order without duplicating information.

Fault Record

Example:

FLT-001284

Ride Pump P 08

Priority: P1

Create Work Order

TICVAI automatically transfers:

- Asset
- location
- fault
- priority
- description
- evidence
- assessment
- operational impact
- safety impact
- SLA
- required skill
- related incident

Work Order

Generated:

WO-002184

Relationship

Maintain:

Fault

↓

Assessment

↓

Corrective Maintenance Request

↓

Work Order

↓

Repair

↓

Testing

↓

Return to Service

Multiple Work Orders

One fault may require multiple work orders.

Example:

Ride Failure

→ Electrical WO

→ Mechanical WO

→ Safety Inspection WO

All remain connected to the original fault.

Board Boundary

The detailed execution of the work order belongs to **Board 4**.

Board 3 owns the fault and corrective-maintenance case.

Screen 8 Fault Investigation and Root Cause Analysis**Purpose**

Identify why an asset failed, especially for critical or recurring failures.

Investigation Record

Capture:

Fault**Asset****Failure Mode****Observed Symptoms****Immediate Cause****Root Cause****Contributing Factors****Root Cause Categories**

Configurable:

- Mechanical
- Electrical
- Network/Communication
- Software
- Configuration
- Wear
- Installation
- Environmental
- Operator Error
- Supplier/Manufacturer
- Maintenance-related
- Unknown

RCA Methods

Support structured methods such as:

5 Whys

and configurable investigation templates.

Example**Problem**

Pump stopped.

Why 1

Motor overheated.

↓

Why 2

Bearing resistance increased.

↓

Why 3

Lubrication insufficient.

↓

Why 4

PM lubrication interval was too long.

Result

Root Cause: Preventive Maintenance Frequency

Follow Up Actions

Change PM Plan

Create Corrective Action

Inspect Similar Assets

Notify Manufacturer

Update Checklist

Screen 9 Repeat Failure and Problem Management**Purpose**

Identify assets that repeatedly fail and prevent maintenance teams from continuously fixing symptoms without addressing the underlying problem.

Repeat Failure Dashboard

Show:

- Assets with repeated faults
- same fault repeated
- failures within 7/30/90 days
- corrective maintenance frequency
- downtime
- maintenance cost

Example**Pump P 08****4 Failures and 30 Days**

Total Downtime: 9h 42m

Maintenance Cost: AED 8,400

Same Failure: 3×

Problem Record

TICVAI can create:

PRB 00042 Repeated Pump Bearing Failure

Linked:

FLT-102

FLT-128

FLT-164

FLT-188

Actions**Start Root Cause Analysis****Inspect Similar Assets****Review PM Plan****Contact Vendor****Recommend Replacement****AI Pattern Detection**

AI may detect:

“Pump model P-X8 has experienced bearing-related failures at three different venues.”

This moves TICVAI from individual fault management toward enterprise reliability management.

Screen 10 Corrective Maintenance Analytics and AI Insights**Purpose**

Provide management with visibility into asset failures, corrective-maintenance performance, operational impact, and emerging reliability risks.

KPI Cards

Display:

- Total Faults
- Critical Faults
- Emergency Events
- Average Response Time
- Average Resolution Time
- Repeat Failure Rate
- Asset Downtime
- SLA Compliance

Failure Analytics

Analyze by:

Tenant**Venue****Asset Category****Asset Type****Manufacturer****Model****Location****Fault Type****Priority****Trend**

Show:

Faults by Week and Month**Top Failing Assets**

Example:

Asset	Faults	Downtime	Cost	Repeat
Pump P-08	4	9.7h	AED 8,400	High
AHU-12	3	6.2h	AED 4,200	Medium

Asset	Faults	Downtime	Cost	Repeat
Gate-04	3	2.1h	AED 1,800	High

AI Insights

Examples:

“Corrective maintenance for Pump P-08 increased 65% during the last 60 days.”

“Assets manufactured by Vendor X have a 2.4× higher failure rate than comparable assets.”

“Most Main Entrance gate failures occur between 17:00 and 20:00.”

“Seven corrective-maintenance cases occurred within 14 days of preventive maintenance completion.”

AI Recommendations

Review PM Frequency

Inspect Similar Assets

Review Vendor

Perform RCA

Consider Replacement

Update Maintenance Checklist

All recommendations remain advisory unless an approved workflow authorizes further action.

Board 3 Shared Configuration Requirements

Authorized administrators shall configure:

- Fault categories
- fault types
- defect classifications
- detection sources
- severity levels
- maintenance priorities
- priority calculation rules
- emergency classifications
- operational impact levels
- safety impact levels
- customer/revenue impact
- response SLA
- resolution SLA
- escalation rules
- emergency response teams
- isolation statuses
- isolation requirements
- corrective-maintenance workflows
- root-cause categories
- RCA templates
- repeat-failure thresholds
- problem-management rules
- mandatory evidence
- notification rules
- approval requirements

Critical Requirement Fault vs Work Order

Softlabs should **not treat the fault and the work order as the same object**.

This distinction is very important.

Fault

Answers:

What is wrong?

Example:

Pump P-08 — abnormal pressure.

Work Order

Answers:

What work needs to be performed?

Example:

Inspect pump, replace bearing and perform operational test.

Therefore:

One Fault

may generate:

One Work Order

or:

Multiple Work Orders

This architecture will allow much stronger analytics later.

Critical Requirement Corrective vs Emergency Maintenance

Not every corrective-maintenance case is an emergency.

Example:

Corrective

Light fixture failed.

Priority P4

Can be repaired tomorrow.

versus:

Emergency

Ride safety sensor failed during operation.

Priority P1

Requires immediate action.

Therefore the same corrective engine should support:

Normal Corrective

and

Emergency Corrective

with different SLA, escalation and operational workflows.

Relationship With Safety Management

If a fault creates a potential safety issue:

Fault

↓

Safety Impact = YES

↓

TICVAI should provide:

Create Safety Incident

The maintenance case and safety incident remain linked.

For example:

Ride restraint sensor failure

may create:

Maintenance Fault FLT 1284

•

Safety Incident SAF 0042

They are related but should not become the same record because maintenance and safety investigations may have different workflows.

Relationship With Device Management

For connected assets, Device Management can automatically create a fault.

Example:

Board 3 Device Monitoring

detects:

Gate Controller GC-04 communication failure.

↓

Device linked to:

AST-00184

↓

TICVAI creates:

Fault FLT 00184

Source:

DEVICE MONITORING

↓

Corrective assessment

↓

Work Order if required.

There should be **one maintenance fault record**, not separate fault records in both modules.

AI Assisted Fault Triage

I recommend including AI assistance during fault reporting.

Example:

Technician enters:

“Pump is making a loud grinding noise and temperature is increasing.”

AI can suggest:

Possible Classification

Mechanical → Bearing / Rotating Component

Suggested Priority

P2 — Critical

Recommended Initial Checks

Bearing / Lubrication / Motor Temperature

But the technician or authorized maintenance user confirms the final classification.

This will make field reporting much faster without giving AI uncontrolled operational authority.

RBAC and Governance

Permissions should distinguish:

Report Fault

Assess Fault

Change Priority

Declare Emergency

Isolate Asset

Create Work Order

Assign Response Team

Create Safety Incident

Perform RCA

Create Problem Record

Close Fault

Override SLA

Return Asset to Service

High-impact actions such as:

Emergency Declaration

Asset Isolation

Priority Downgrade

Safety Critical Closure

can require approval depending on tenant policy.

Every critical action must record:

Who | What | When | Previous Value | New Value | Reason

Board 3 Acceptance Criteria

Board 3 is accepted when TICVAI can successfully:

Detect/Report Fault → Assess → Classify → Prioritize → Escalate → Isolate if Required → Generate Corrective Work → Investigate → Identify Root Cause → Detect Repeat Failure → Analyze

asset faults and emergency maintenance events.

The solution must support normal corrective maintenance and emergency response while maintaining direct integration with preventive maintenance, work-order execution, asset management, safety management, and Device Management.

Most importantly, TICVAI must preserve the complete chain:

Asset → Fault → Assessment → Impact → Work Order → Repair → Root Cause → Return to Service → Reliability History

Board Boundary

Board 1 — Asset identity & lifecycle

Board 2 — Preventive maintenance planning

Board 3 — Corrective & emergency maintenance

Board 4 — Work-order execution & field operations

Board 5 — Safety & compliance

Board 6 — Spare parts & maintenance inventory

Board 7 — Procurement & maintenance resources

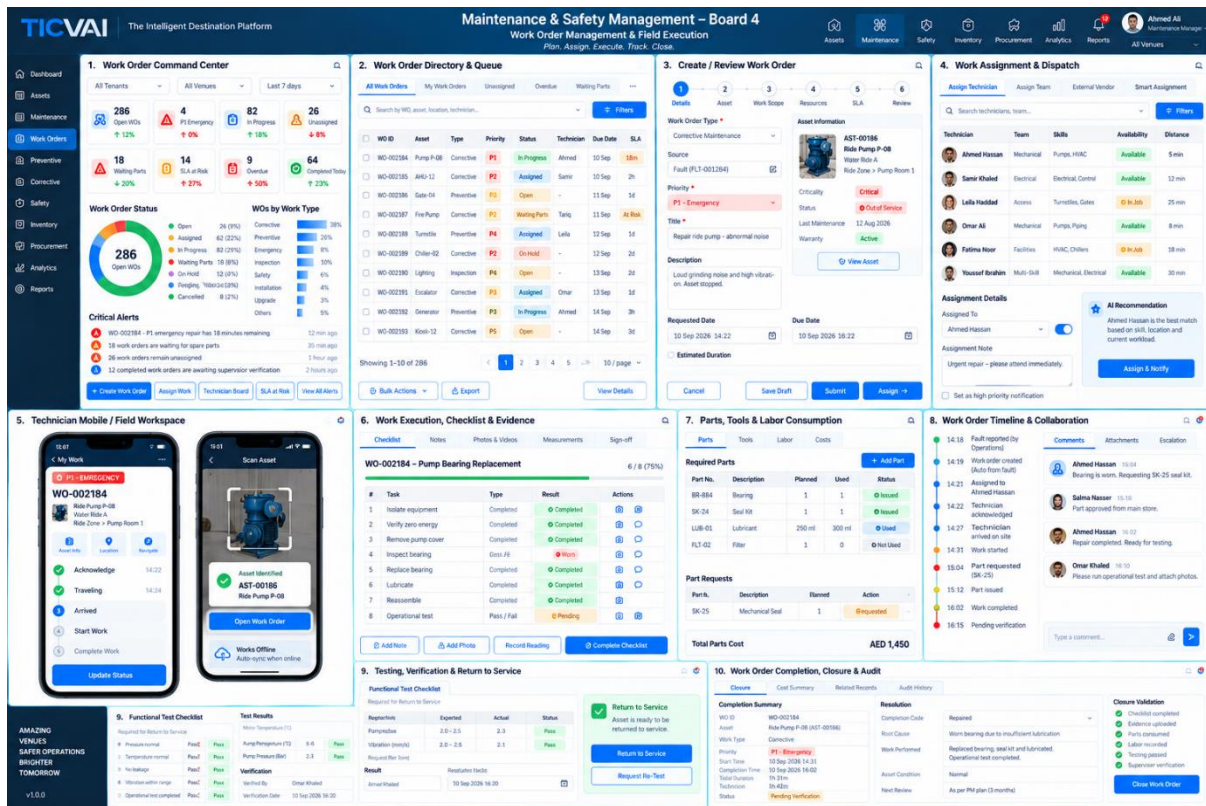
Board 8 — Maintenance intelligence & reporting

Board 3 = 10 screens.

Board 4 Work Order Management and Field Execution

Board Objective: Provide TICVAI with the central execution engine for all maintenance work orders—from creation and assignment through technician execution, parts consumption, testing, approvals, closure, and complete audit history.

Board 4 is where maintenance becomes **actual executable work**.



The core principle is:

Every maintenance activity that requires someone to perform physical work should ultimately be controlled through a governed TICVAI Work Order.

A work order may originate from:

Preventive Maintenance | Corrective Maintenance | Emergency Maintenance | Safety Inspection | Device Fault | Manual Request | Asset Inspection | AI Recommendation | API / External System

The architecture must maintain **one Work Order Engine** rather than creating separate work-order systems for each TICVAI module.

Screen 1 Work Order Command Center

Purpose

Provide maintenance managers, supervisors, technicians, and operations teams with a real-time overview of all maintenance work.

KPI Cards

Display:

- Open Work Orders
- Emergency / P1
- Assigned
- Unassigned
- In Progress
- Waiting for Parts
- Waiting for Approval

- SLA at Risk
- Overdue
- Completed Today

Example:

KPI	Value
Open Work Orders	286
P1 Emergency	4
Unassigned	26
In Progress	82
Waiting Parts	18
SLA at Risk	14
Overdue	9
Completed Today	64

Work Order Status Overview

Support:

- ☐ Draft
- ☐ Open
- ☐ Assigned
- ☒ In Progress
- ☒ On Hold
- ☒ Waiting Parts
- ☒ Waiting Vendor
- ☐ Completed
- ☐ Pending Verification
- ☐ Closed
- ☒ Cancelled

Breakdown

Analyze by:

Tenant | Venue | Location | Asset Category | Work Type | Priority | Team | Technician

Critical Attention Panel

Examples:

- ☒ WO-002184 — P1 emergency repair has 18 minutes remaining before SLA breach.
- ☒ 18 work orders are waiting for spare parts.
- ☐ 26 work orders remain unassigned.
- ☐ 12 completed work orders are awaiting supervisor verification.

Quick Actions

+ Create Work Order

Assign Work**Technician Board****SLA at Risk****Waiting Parts****Calendar****Screen 2 Work Order Directory and Queue****Purpose**

Provide the central searchable operational queue for every maintenance work order.

Work Order Table

WO	Asset	Type	Priority	Status	Technician	SLA
WO-002184	Pump P-08	Emergency	P1	In Progress	Ahmed	18m
WO-002185	AHU-12	Corrective	P2	Assigned	Samir	2h
WO-002186	Gate-04	Preventive	P3	Open	—	1d
WO-002187	Fire Pump	Inspection	P2	Waiting Parts	Tariq	At Risk

Search

Search by:

- Work Order ID
- Asset ID
- Asset Name
- Fault ID
- Technician
- Location
- Serial number
- QR/barcode

Filters

Support:

- Tenant
- Venue
- Location
- Work type
- Priority
- Status
- Asset category
- Technician
- Team
- Vendor
- SLA
- Date
- Source

Saved Views

Examples:

My Work Orders**P1/P2****Unassigned****Due Today****Waiting Parts****Pending Approval****Overdue****Bulk Actions**

Authorized users can:

Assign**Reassign****Change Priority****Reschedule****Export****Cancel****Screen 3 Create and Review Work Order****Purpose**

Provide a governed form for manually creating a work order or reviewing one generated automatically.

Work Order Information

Capture:

- Work Order ID
- Work Order Type
- Source
- Asset
- Location
- Problem / Work Description
- Priority
- Requested Date
- Due Date
- SLA
- Estimated Duration

Work Order Types

Configurable:

Preventive**Corrective****Emergency****Inspection****Safety****Installation****Replacement****Upgrade****General Maintenance****Source Record**

Example:

Source: Corrective Fault

Fault: FLT-001284

The user can open the originating record.

Asset Context

Automatically display:

Asset Criticality

Operational Status

Location

Last Maintenance

Open Faults

Warranty

Work Scope

Capture:

- Required activities
- checklist
- required skill
- estimated labor
- tools
- parts
- permits
- safety requirements

Actions

Save Draft

Submit

Approve

Assign

depending on workflow and permissions.

Screen 4 Work Assignment and Dispatch

Purpose

Assign the right technician, team, or external maintenance provider to the work.

Assignment Panel

Display available:

Internal Technicians

Maintenance Teams

Specialists

External Vendors

Technician Information

Show:

- Name
- Team
- Current location
- Shift
- Availability
- Skill

- Certification
- Current workload
- Open P1/P2 jobs

Example**Ahmed Hassan****Mechanical Team**

Availability:

 Available

Current Jobs:

2

Skill:

Pump / Mechanical

Certification:

Valid

Distance:

Zone B – 5 min

Smart Assignment

TICVAI can recommend technicians using:

Required Skill

-

Certification

-

Location

-

Availability

-

Workload

-

Priority**AI Recommendation**

“Ahmed Hassan is the recommended technician based on mechanical certification, current venue location, and workload.”

Dispatch

Once assigned:

Technician notified

↓

Work appears in technician queue

↓

Technician acknowledges**Assignment SLA**

Track:

Created → Assigned → Acknowledged → Started

This allows management to identify dispatch delays separately from repair delays.

Screen 5 Technician Mobile and Field Workspace

Purpose

Provide technicians with a simple field interface to perform maintenance without requiring them to navigate the full administration system.

This screen should be designed **mobile-first**, although desktop/tablet versions should also be available.

Header

WO-002184

P1 — EMERGENCY

Ride Pump P 08

Asset Information

Display:

- Asset
- location
- QR
- fault
- priority
- safety instructions
- expected completion

Technician Actions

Acknowledge

Traveling

Arrived

Start Work

Pause

Resume

Complete Work

Field Information

Technician can:

- Add notes
- take photos
- upload videos
- enter readings
- complete checklist
- record diagnosis
- record repair
- request parts
- request assistance

Asset QR

Technician may scan the asset before beginning.

TICVAI verifies:

✓ Correct asset confirmed.

This is particularly useful in locations containing many similar assets.

Offline Capability

The field workspace should support controlled offline operation where practical.

Technician can:

Open cached WO → Perform checklist → Capture information → Synchronize when connectivity returns

Conflict and synchronization rules must be defined.

Screen 6 Work Execution Checklist and Evidence

Purpose

Control exactly what work was performed and capture evidence of maintenance execution.

Execution Checklist

Example:

Pump Bearing Replacement

- ☒ Isolate equipment
- ☒ Verify zero energy
- ☒ Remove pump cover
- ☒ Inspect bearing
- ☒ Replace bearing
- ☒ Lubricate
- ☒ Reassemble
- ☐ Operational test
- ☐ Supervisor verification

Task Types

Support:

- Checkbox
- Pass / Fail
- Numeric reading
- Text
- Photo
- Video
- Signature
- Measurement
- Attachment

Before and After Evidence

Allow:

Before Photo

and

After Photo

Measurements

Example:

Motor Temperature Before

82°C

Motor Temperature After

54°C

Failed Checklist Item

If:

Operational Test = FAIL

TICVAI should prevent normal completion and offer:

Continue Repair**Escalate****Create Additional Work****Request Specialist****Mandatory Evidence**

Critical tasks may require:

Photo + Reading + Technician Signature

before they can be marked complete.

Screen 7 Parts Tools and Labor Consumption**Purpose**

Capture the actual resources consumed during maintenance.

This screen connects directly with **Board 6 — Spare Parts & Maintenance Inventory**.

Parts Required

Example:

Part	Planned	Used	Status
Bearing BR-884	1	1	Issued
Seal Kit SK-24	1	1	Issued
Lubricant	250ml	300ml	Used

Actions**Reserve Part****Issue Part****Consume****Return Unused****Request Part****Insufficient Stock**

Example:

● Seal Kit SK-24 is unavailable at Dubai Park store.

Actions:

Check Other Store**Request Transfer****Create Procurement Request****Tools**

Record required/special tools.

Example:

Torque Wrench**Diagnostic Meter****Labor**

Capture:

Technician**Start****Finish****Active Labor Time****Overtime****Example**

Ahmed:

2h 18m

Samir:

1h 42m

Total:

4 labor hours

Maintenance Cost

Calculate where permitted:

Labor

-

Parts

-

External Vendor

-

Other Cost

=

Actual Work Order Cost**Screen 8 Work Order Timeline Collaboration and Escalation****Purpose**

Provide a complete real-time history and collaboration workspace for the work order.

Timeline

Example:

14:18

Fault reported

↓

14:19

WO generated

↓

14:21

Ahmed assigned

↓

14:22

Technician acknowledged

↓

14:27

Technician arrived

↓

14:31

Work started

↓

15:04

Part requested

↓

15:12

Part issued

↓

16:02

Repair completed

Activity Feed

Users can add:

- Comments
- technician notes
- supervisor instructions
- photos
- attachments
- vendor updates

Mention

Example:

@MaintenanceSupervisor please review the bearing damage.

Escalation

Authorized users can:

Escalate Priority**Request Specialist****Request Vendor****Notify Operations****Create Safety Incident****On Hold**

Work order can be placed:

On Hold Waiting Parts**On Hold Waiting Vendor****On Hold Operational Access****On Hold Approval**

The reason and duration must be tracked.

SLA Treatment

Whether an on-hold period pauses SLA must be configurable.

This is important because some clients may allow SLA pause for vendor/parts delays while others may not.

Screen 9 Testing Verification and Return to Service**Purpose**

Ensure repaired assets are not returned to operation simply because the technician has finished physical work.

This is especially important for **rides, gates, safety equipment, electrical systems, mechanical systems, and other critical assets**.**Repair Completed**

Technician selects:

Repair Complete

This does **not necessarily mean the Work Order is closed.**

Post Maintenance Testing

Configure tests by asset category.

Example:

Turnstile

- ☒ Controller online
- ☒ Reader operational
- ☒ Entry command successful
- ☒ Gate opened
- ☒ Gate feedback received
- ☒ Emergency function tested

Pump

- ☒ Pressure normal
- ☒ Temperature normal
- ☒ No leakage
- ☒ Vibration within range
- ☒ Operational test completed

Test Result

☒ **PASS**

or

☐ **FAIL**

Verification

Depending on asset criticality:

Technician Self Verification

or

Supervisor Verification

or

Safety Verification**Return to Service**

Authorized user selects:

Return Asset to Service

Record:

- Verified by
- verification time
- test results
- asset condition
- restrictions
- comments

Asset status changes:

Under Maintenance

→

Operational**Failed Verification**

If test fails:

Reopen Work

and maintain the same work-order history.

Screen 10 Work Order Completion Closure and Audit**Purpose**

Provide a governed closure process and complete permanent maintenance record.

Completion Summary

Display:

WO: WO-002184

Asset: Pump P-08

Type: Emergency Corrective

Priority: P1

Work Performed

Technician enters:

Replaced worn bearing, replaced seal, lubricated assembly and completed operational testing.

Resolution Code

Configurable:

- Repaired
- Replaced Component
- Temporary Repair
- No Fault Found
- Adjustment
- Configuration Change
- Vendor Repair
- Asset Replacement
- Other

Final Information

Capture:

- Root cause
- work performed
- parts used
- labor
- downtime
- actual cost
- test results
- evidence
- technician signature
- supervisor approval

Closure Validation

Before closure TICVAI checks:

- ✓ Mandatory checklist complete
- ✓ Required evidence uploaded

- ✓ Parts consumption recorded
- ✓ Labor recorded
- ✓ Testing passed
- ✓ Required approvals completed

Close Work Order

Status:

Completed

↓

Verified

↓

Closed

Audit History

Maintain:

Who | Action | Previous Value | New Value | Date/Time | Reason

The closed work order becomes part of the permanent **Asset Maintenance History** from Board 1.

Board 4 Shared Configuration Requirements

Authorized administrators shall configure:

- Work-order types
- work-order numbering
- source types
- statuses
- status transitions
- priorities
- default SLA
- assignment rules
- dispatch rules
- technician skills
- required certifications
- checklist templates
- mandatory tasks
- evidence requirements
- mobile permissions
- offline rules
- parts reservation rules
- labor recording rules
- cost calculations
- on-hold reason codes
- SLA pause rules
- escalation rules
- completion codes
- resolution codes
- testing templates
- verification rules

- return-to-service permissions
- closure requirements
- approval workflows
- cancellation reasons
- reopen rules

Work Order State Machine

I strongly recommend Softlabs build the work-order lifecycle as a configurable **state machine**, rather than hardcoding a simple Open/Closed status.

A typical lifecycle should be:

Draft

↓

Open

↓

Assigned

↓

Acknowledged

↓

In Progress

↓

Work Completed

↓

Testing

↓

Pending Verification

↓

Returned to Service

↓

Closed

Alternative branches:

In Progress → Waiting Parts

In Progress → Waiting Vendor

In Progress → On Hold

Testing Failed → In Progress

Completed → Reopened

This gives TICVAI much stronger enterprise flexibility.

Critical Architecture One Central Work Order Engine

This is one of the most important architectural requirements in Module 17.

TICVAI should **not** create:

- Preventive Work Order System
- Corrective Work Order System
- Device Work Order System
- Safety Work Order System
- Inspection Work Order System

Instead:

TICVAI Central Work Order Engine

receives work from:

Preventive Maintenance

↓

Corrective Maintenance

↓

Emergency Maintenance

↓

Safety and Compliance

↓

Device Management

↓

Manual Request

↓

API and External System

All produce the same core:

WO-XXXXXX

The **source and work type** determine the workflow, checklist, priority, SLA, and approval requirements.

Relationship With Boards 1 3**Board 1**

Provides:

Asset**Board 2**

Provides:

Preventive Maintenance Requirement**Board 3**

Provides:

Fault and Corrective Requirement**Board 4**

Provides:

Execution

Therefore:

Asset

↓

PM Plan OR Fault

↓

Work Order

↓

Technician

↓

Parts Labor Checklist

↓

Testing

↓

Return to Service

↓

Asset History

This gives TICVAI full traceability.

Relationship With Board 5 Safety

If a work order involves safety-sensitive work, Board 4 should consume safety requirements from Board 5.

Example:

Electrical Maintenance

Before **Start Work**, require:

Safety Permit**Isolation Confirmation****Required PPE****Authorized Technician****Ride Maintenance**

Before **Return to Service**, require:

Safety Inspection

and potentially:

Supervisor and Safety Approval

The Work Order remains the execution record, while Board 5 remains the safety/compliance authority.

Relationship With Board 6 Spare Parts

The Work Order should not maintain a separate inventory.

Instead:

WO requests part

↓

Board 6 reserves stock

↓

Store issues part

↓

Technician consumes part

↓

Inventory reduced

↓

Cost posted to Work Order

If stock is unavailable:

WO

↓

Part Request

↓

Transfer or Procurement

This provides one inventory source of truth.

Relationship With Device Management

For connected equipment:

Device Monitoring

↓

Detects fault

↓

Maintenance Fault

↓

Work Order

↓

Technician repair

↓

Device health test

↓

Return to Service

The technician can access relevant Device Management information from the Work Order, such as:

- Connectivity
- Firmware
- Last heartbeat
- Device logs
- Configuration
- Diagnostics

But Board 4 remains the **maintenance execution system**.

Mobile First Technician Experience

For technicians, I recommend Softlabs avoid reproducing the full desktop interface on mobile.

The field experience should focus on:

MY WORK

↓

Open WO

↓

Scan Asset

↓

Start

↓

Checklist

↓

Parts

↓

Evidence

↓

Test

↓

Complete

Technicians should be able to execute common tasks with very few taps.

This will be important for large venues where technicians are moving continuously between assets.

Offline Field Operations

For TICVAI's target venues, Board 4 should support an offline-capable technician experience.

A technician may enter:

Basement**Mechanical Room****Back of House****Ride Infrastructure Area**

where network coverage is weak.

Therefore:

Assigned WO

↓

Securely cached on device

↓

Technician performs work offline

↓

Evidence and checklist stored locally

↓

Connectivity restored

↓

Synchronize

The system must maintain conflict handling and a clear synchronization state.

For high-risk actions such as final safety approval, tenants may configure **online verification as mandatory**.

SLA Measurement

TICVAI should measure different time components separately.

For example:

Fault Reported

→ 3 min →

WO Created

→ 4 min →

Assigned

→ 2 min →

Acknowledged

→ 8 min →

Technician Arrived

→ 1h 20m →

Repair Completed

→ 12 min →

Verified

This provides:

Assignment Time**Response Time****Travel Time****Repair Time****Waiting Time****Verification Time****Total Resolution Time**

This is much more valuable than measuring only total WO duration.

RBAC and Governance

Permissions should distinguish:

Create Work Order

Assign

Reassign

Accept Work

Start Work

Pause Work

Request Parts

Consume Parts

Add Evidence

Change Priority

Place On Hold

Escalate

Complete Work

Perform Testing

Verify

Return to Service

Close

Reopen

Cancel

For critical assets, the same person who performs the repair may optionally be prevented from giving the final return-to-service approval.

This supports **segregation of duties**.

Board 4 Acceptance Criteria

Board 4 is accepted when TICVAI can successfully manage:

Create/Receive WO → Prioritize → Assign → Dispatch → Acknowledge → Execute → Record Checklist/Evidence → Consume Parts/Labor → Test → Verify → Return to Service → Close → Audit

for maintenance activities across all tenants and venues.

The solution must support preventive, corrective, emergency, inspection, safety, device-generated, manual, and externally generated work orders through **one centralized Work Order Engine**.

The complete traceability chain must be preserved:

Source → Asset → Work Order → Technician → Work Performed → Parts → Labor → Evidence → Testing → Approval → Return to Service → Closure → Asset History

Board Boundary

Board 1 — Asset identity & lifecycle

Board 2 — Preventive maintenance planning

Board 3 — Corrective & emergency maintenance

Board 4 — Work-order execution & field operations

Board 5 — Safety & compliance

Board 6 — Spare parts & maintenance inventory

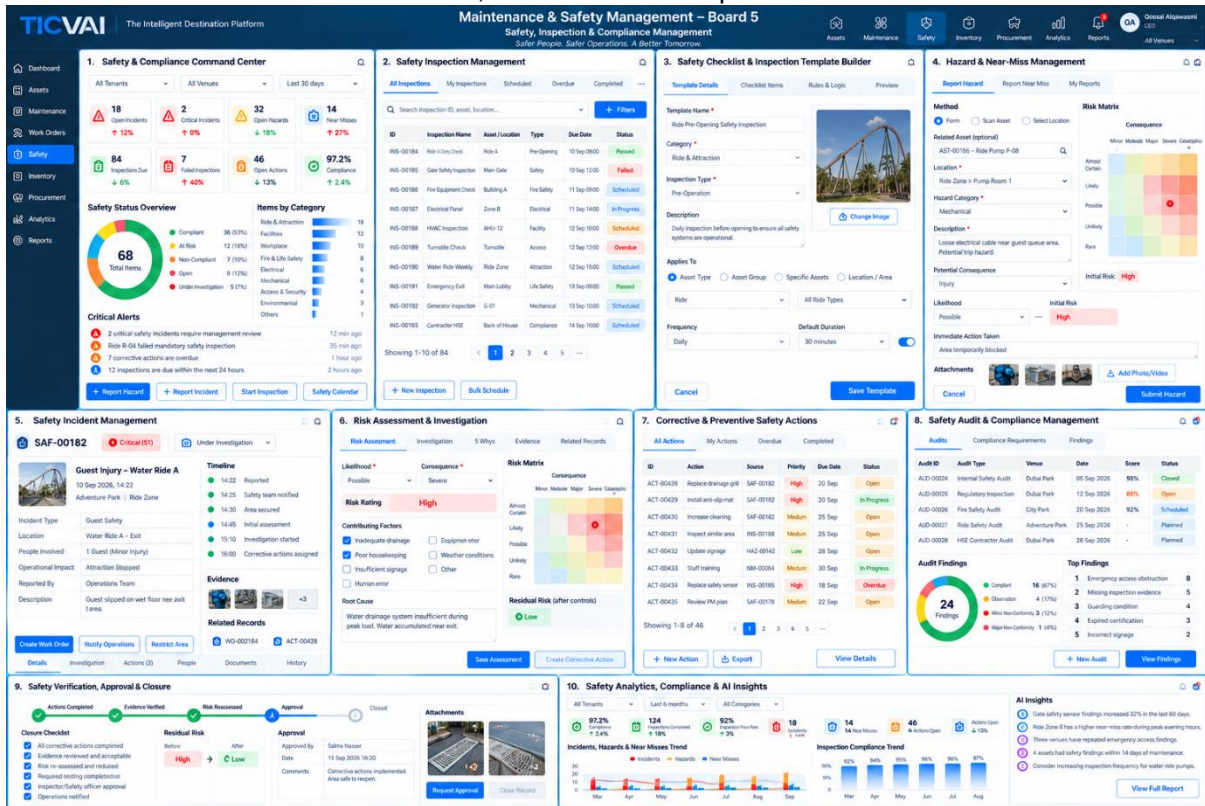
Board 7 — Procurement & maintenance resources

Board 8 — Maintenance intelligence & reporting

Board 4 = 10 screens.

Board 5 Safety Inspection and Compliance Management

Board Objective: Provide TICVAI with a centralized safety and compliance engine for managing **safety inspections, configurable checklists, hazards, incidents, near misses, corrective actions, safety audits, compliance requirements, evidence, and closure** across all tenants, venues, assets, attractions, facilities, and maintenance operations.



The core principle is:

Safety must not be treated as only a maintenance checklist. TICVAI must provide a governed process from identification of a safety risk through investigation, corrective action, verification, and final closure.

Board 5 should also integrate tightly with Assets, Preventive Maintenance, Corrective Maintenance and Work Orders without duplicating their records.

Screen 1 Safety and Compliance Command Center

Purpose

Provide safety managers, venue management, maintenance teams and authorized executives with a real-time overview of safety performance and compliance.

KPI Cards

Display:

- Open Safety Incidents
- Critical Incidents
- Open Hazards
- Near Misses
- Inspections Due
- Inspections Overdue
- Failed Inspections
- Open Corrective Actions
- Compliance %

- Safety Actions Overdue

Example:

KPI	Value
Open Incidents	18
Critical Incidents	2
Open Hazards	32
Near Misses	14
Inspections Due	84
Failed Inspections	7
Open Actions	46
Compliance	97.2%

Safety Status Overview

Show:

- ☒ **Compliant**
- ☐ **Scheduled**
- ☐ **Attention Required**
- ☒ **At Risk**
- ☒ **Non-Compliant / Critical**

Breakdown

Analyze by:

Tenant | Venue | Location | Asset | Safety Category | Severity | Inspection Type | Status

Critical Attention Panel

Examples:

- ☒ Two critical safety incidents require management review.
- ☒ Ride R-04 failed a mandatory safety inspection.
- ☒ Seven corrective safety actions are overdue.

⚠ 12 safety inspections are due within the next 24 hours.

Quick Actions

+ Report Hazard

+ Report Incident

Start Inspection

Safety Calendar

Corrective Actions

Compliance Status

Screen 2 Safety Inspection Management

Purpose

Create, schedule, assign and manage recurring or ad-hoc safety inspections.

Inspection Types

Support configurable types such as:

- Daily Opening Inspection

- Pre-Operation Inspection
- Ride Safety Inspection
- Equipment Inspection
- Fire & Life Safety
- Electrical Safety
- Facility Inspection
- Workplace Safety
- Emergency Equipment Inspection
- Contractor Inspection
- Regulatory Inspection
- Post-Maintenance Inspection
- Ad-hoc Inspection

Inspection Directory

Inspection	Asset/Area	Type	Due	Inspector	Status
INS-00184	Ride A	Pre-Opening	10 Sep 08:00	Ahmed	Passed
INS-00185	Gate 04	Safety	10 Sep 12:00	Omar	Failed
INS-00186	Zone B	Fire Safety	11 Sep	Sara	Scheduled

Inspection Scope

Inspection can apply to:

Asset

Asset Group

Attraction

Area

Building

Venue

Scheduling

Support:

One Time

Daily

Weekly

Monthly

Annual

Custom

and where required:

Before Opening

After Maintenance

Before Return to Service

Assignment

Assign:

Inspector**Safety Team****Maintenance Team****External Inspector****Screen 3 Safety Checklist and Inspection Template Builder****Purpose**

Allow administrators and safety managers to build reusable safety inspection templates without hardcoding each inspection.

Template Example**Ride Pre Opening Safety Inspection**

1. Restraint system condition
2. Emergency stop function
3. Safety sensor status
4. Guarding condition
5. Operating area clear
6. Emergency access clear
7. Communication equipment operational
8. Warning signage visible
9. Required documentation available
10. Final operational safety confirmation

Checklist Response Types

Support:

Pass and Fail**Yes and No****Compliant and Non Compliant****Numeric Reading****Measurement****Text****Photo****Video****Signature****Attachment****Mandatory Evidence**

Example:

Emergency Stop Test

Result:

PASS

Required:



Photo

or:

Inspector Signature**Conditional Logic**

Example:

If:

Guard Condition = FAIL

then automatically require:

Failure Description

Photo Evidence

Severity

Corrective Action

Critical Items

Certain checklist items should be configurable as:

SAFETY CRITICAL

Failure may automatically:

- Fail the inspection
- restrict return to service
- generate a hazard
- generate corrective action
- notify safety management

depending on configured policy.

Screen 4 Hazard and Near Miss Management

Purpose

Allow staff to report unsafe conditions and near misses before they become actual incidents.

This is important because **hazards, near misses and incidents are different records**.

Report Hazard

Capture:

- Hazard ID
- category
- description
- tenant
- venue
- location
- related asset
- detected by
- date/time
- potential consequence
- likelihood
- evidence

Example

HAZ-00142

Loose electrical cable near Guest Queue

Location:

Ride Zone → Queue Entrance

Potential Impact:

Trip / Electrical Hazard

Near Miss

Example:

Guest nearly slipped due to water leakage, but no injury occurred.

Create:

NM-00084**Quick Field Reporting**

The field experience should be simple:

Select and Scan Location or Asset

↓

Hazard and Near Miss

↓

Photo

↓

Short Description

↓

Submit**Immediate Action**

Reporter can record:

Area temporarily blocked.

or:

Supervisor notified.

Screen 5 Safety Incident Management**Purpose**

Manage actual safety incidents through a controlled lifecycle.

Incident Record

Capture:

- Incident ID
- incident type
- date/time
- tenant
- venue
- exact location
- related asset
- people involved
- description
- severity
- operational impact
- immediate actions
- evidence
- reported by

Incident Categories

Configurable:

- Guest Safety
- Employee Safety
- Contractor Safety
- Equipment Safety
- Ride/Attraction
- Electrical

- Fire
- Slip/Trip/Fall
- Mechanical
- Facility
- Environmental
- Other

Incident Severity

Recommended:

S1 Critical

S2 Major

S3 Moderate

S4 Minor

S5 Observation

Immediate Actions

Depending on permission:

Notify Safety Team

Notify Operations

Create Maintenance Fault

Create Work Order

Isolate Asset

Restrict Area

Start Investigation

Timeline

Maintain:

Reported → Assessed → Controlled → Investigated → Actions Assigned → Verified → Closed

Screen 6 Safety Risk Assessment and Investigation

Purpose

Evaluate safety risk consistently and investigate why an incident, hazard, or failed inspection occurred.

Risk Matrix

I recommend TICVAI support a configurable:

Likelihood Consequence

matrix.

Example:

	Minor	Moderate	Major	Severe	Catastrophic
Rare	Low	Low	Medium	Medium	High
Unlikely	Low	Medium	Medium	High	High
Possible	Medium	Medium	High	High	Critical
Likely	Medium	High	High	Critical	Critical
Almost Certain	High	High	Critical	Critical	Critical

Risk Result

Example:

Likelihood: Possible

Consequence: Severe

Result:

● **HIGH RISK**

Investigation

Capture:

- What happened?
- Where?
- When?
- Who was involved?
- Immediate cause
- root cause
- contributing factors
- existing controls
- failed controls
- evidence

RCA

Support:

5 Whys

Contributing Factors

Cause Classification

Evidence

Allow:

Photos

Videos

Documents

Witness Statements

Inspection Results

Maintenance History

Device Logs

Screen 7 Corrective and Preventive Safety Actions

Purpose

Ensure safety findings lead to accountable actions rather than remaining only as reports.

Action Record

Example:

ACT-00428

Source: Safety Incident SAF-00182

Action: Replace damaged gate safety sensor.

Capture

- Action
- owner
- responsible department
- priority
- due date
- status

- evidence required
- verification required

Action Types

Immediate Correction

Corrective Action

Preventive Action

Process Change

Training

Inspection

Maintenance

Asset Replacement

Action Status

Open

- Assigned
- In Progress
- Completed
- Pending Verification
- Verified
- Closed

Work Order Integration

If physical maintenance is required:

Safety Action

↓

Create Work Order

↓

WO-002248

The safety action remains linked to the work order.

Overdue Action

Example:

● ACT-00428 is 3 days overdue.

Escalate according to configured policy.

Screen 8 Safety Audit and Compliance Management

Purpose

Manage formal safety audits and compliance requirements across venues and operational areas.

Compliance Library

Support configurable requirements such as:

- Internal Safety Policy
- Venue Safety Standard
- Manufacturer Requirements
- Operating Procedures
- Regulatory Requirements
- Certification Requirements
- Insurance Requirements
- Contractual Safety Requirements

TICVAI should not hardcode specific regulations because requirements differ between jurisdictions and clients.

Audit Record

Capture:

- Audit ID
- audit type
- scope
- venue
- auditor
- audit date
- applicable requirements
- findings
- evidence
- score
- status

Audit Findings

Classify:



Compliant



Observation



Minor Non-Conformity



Major Non-Conformity

Finding

Example:

FIND-00128

Requirement: Emergency Exit Inspection

Result: Non-Compliant

Finding: Emergency access partially obstructed.

Actions

Create Corrective Action

Assign Owner

Set Due Date

Upload Evidence

Request Verification

Compliance Score

Example:

Venue Safety Compliance

97.2

Screen 9 Safety Verification Approval and Closure

Purpose

Ensure safety records cannot be closed merely because someone marks an action as complete.

Closure Review

Display:

Incident and Hazard and Finding



Required Actions**Related Work Orders****Evidence****Verification****Example****SAF-00182**

Corrective Actions:

4 / 4 Completed

Related Work Orders:

2 / 2 Closed

Required Evidence:

Complete

Verification Checklist

- ☒ Hazard eliminated
- ☒ Corrective work completed
- ☒ Evidence reviewed
- ☒ Required testing completed
- ☒ Risk reassessed
- ☒ Residual risk acceptable

Residual Risk

Before:

☒ **HIGH**

After Controls:

☒ **LOW**

Verification Roles

Depending on configuration:

Supervisor

Safety Officer

Maintenance Manager

Independent Inspector

Segregation of Duties

For critical incidents:

The person who completed the corrective action should optionally **not be allowed to provide final safety verification.**

Closure

Status:

Pending Verification**Verified****CLOSED**

All records remain permanently auditable.

Screen 10 Safety Analytics Compliance and AI Insights

Purpose

Provide management with consolidated intelligence on safety performance, recurring risks, inspection compliance, incidents and corrective actions.

KPI Cards

Display:

- Safety Compliance %
- Inspections Completed
- Inspection Pass Rate
- Open Hazards
- Near Misses
- Incidents
- Critical Incidents
- Corrective Actions Overdue
- Average Closure Time
- Repeat Safety Findings

Trend Analysis

Show:

Incidents by Month

Hazards by Month

Near Miss Reporting

Inspection Pass Rate

Corrective Action Compliance

Analysis

Filter by:

Tenant

Venue

Area

Asset Category

Asset

Incident Type

Risk Level

Inspection Type

Department

Risk Heatmap

Show:

Likelihood Consequence

with number of open risks in each cell.

Repeat Findings

Example:

Finding	Occurrences	Venues	Trend
Emergency Access Obstruction	8	3	↑

Finding	Occurrences	Venues	Trend
Gate Safety Sensor	6	2	↑
Missing Inspection Evidence	5	4	↓

AI Insights

Examples:

“Gate safety-sensor findings have increased 32% during the last 60 days.”

“Three venues have repeated emergency-access findings despite previous corrective actions.”

“Ride Zone B has a higher near-miss rate during peak evening operations.”

“Four assets experienced safety findings within 14 days of corrective maintenance.”

AI Recommendations

Review Maintenance Procedure

Increase Inspection Frequency

Inspect Similar Assets

Review Training

Create Safety Audit

Review Asset Reliability

AI should recommend actions but must not independently close incidents, change compliance status, or authorize return to service.

Board 5 Shared Configuration Requirements

Authorized administrators should be able to configure:

- Safety categories
- incident categories
- hazard categories
- near-miss categories
- inspection types
- inspection templates
- checklist templates
- checklist response types
- critical checklist items
- mandatory evidence
- inspection frequencies
- inspection assignment rules
- severity levels
- likelihood levels
- consequence levels
- risk matrices
- risk thresholds
- escalation rules
- incident workflows
- hazard workflows
- near-miss workflows
- RCA categories
- corrective-action types

- action priorities
- due-date rules
- compliance requirements
- audit types
- finding classifications
- approval requirements
- verification requirements
- segregation-of-duty rules
- closure requirements
- notification rules
- retention policies

Critical Architecture Safety Objects Must Remain Separate

Softlabs should **not create one generic “Safety Record”** for everything.

TICVAI should distinguish at least:

Hazard

A dangerous condition exists.

Near Miss

Something happened that could have caused harm, but did not.

Incident

A safety event actually occurred.

Inspection

A structured check was performed.

Audit

Compliance was formally assessed.

Finding

An inspection/audit identified a problem.

Corrective Action

Something must be done to eliminate or reduce the problem.

These records should be **linked**, but not merged.

Example:

Inspection INS 00184

↓

Failed Item

↓

Finding FIND 00128

↓

Hazard HAZ 0042

↓

Corrective Action ACT 00428

↓

Work Order WO 002248

↓

Repair Completed

↓

Safety Verification

↓

Finding Closed

This gives TICVAI enterprise-grade traceability.

Critical Architecture Safety vs Maintenance

This distinction is also important.

Maintenance asks:

Is the asset technically working?

Safety asks:

Is the asset/operation safe and compliant?

These are not always the same thing.

For example:

A ride may technically operate correctly after a repair, but still require:

Safety Inspection

and:

Safety Approval

before returning to guest operation.

Therefore:

Work Completed

does not automatically mean:

Safe to Operate

for assets where a safety-verification policy applies.

Relationship With Board 4 Work Orders

Board 5 should not create another maintenance execution system.

If a safety issue requires physical work:

Safety Finding

↓

Corrective Action

↓

Board 4 Work Order

↓

Maintenance Execution

↓

Testing

↓

Work Order Completed

↓

Board 5 Safety Verification

↓

Safety Record Closed

This preserves one Work Order Engine.

Relationship With Board 1 Assets

Safety records should link directly to the Asset 360° profile.

Example:

Ride Pump P 08

Asset Profile should eventually show:

Maintenance**Work Orders****Safety**

Within Safety:

- Open hazards
- incidents
- inspections
- failed inspections
- audit findings
- corrective actions

This creates a complete asset risk history.

Relationship With Device Management

Connected devices can contribute safety information.

Example:

Gate Controller

↓

Device monitoring detects:

Safety Input Failure

↓

Device Management Alert

↓

Maintenance Fault

and where configured:

↓

Safety Hazard

The Device Manager remains responsible for technical telemetry.

Board 5 remains responsible for:

Safety Risk → Investigation → Action → Compliance → Verification

Critical Requirement QR Based Safety Inspection

Since Board 1 already provides QR/barcode identification, TICVAI should use the same asset identifier for field inspections.

Technician or safety inspector:

Scans Asset QR

↓

Asset identified

↓

Select:

Start Safety Inspection

↓

Correct checklist automatically loaded based on:

Asset Type Venue Inspection Type

This is much better than creating separate QR labels for maintenance and safety.

Safety Permit and Permit to Work Readiness

I recommend that the architecture of Board 5 be designed so TICVAI can support **Permit-to-Work** workflows, even if advanced permit functionality is phased later.

Examples:

- Electrical Work
- Hot Work
- Work at Height
- Confined Space
- Lockout/Tagout
- Restricted Area Work

A Work Order could then require:

Valid Permit

↓

before:

START WORK

This should be configurable by tenant, asset type and work type rather than hardcoded.

Risk Scoring

TICVAI should maintain both:

Initial Risk

Risk before corrective controls.

and:

Residual Risk

Risk after corrective controls.

Example:

Initial Risk

● HIGH

↓

Corrective Actions

↓

Residual Risk

● LOW

This allows management to understand whether corrective actions actually reduced risk.

Mobile Safety Experience

Like Board 4, the field safety experience should be mobile-first.

The common workflow should be extremely simple:

REPORT

Hazard | Near Miss | Incident

↓

Scan Asset and Select Location

↓

Photo

↓

Description

↓

Severity and Immediate Action

↓

SUBMIT

For inspections:

INSPECT**Scan Asset**

↓

Load Checklist

↓

Pass and Fail

↓

Evidence

↓

Sign

↓

SUBMIT

This is particularly important because complicated safety forms discourage field reporting.

AI Safety Governance

AI can assist with:

- Classification
- Pattern detection
- repeated findings
- risk trend identification
- suggested investigation areas
- suggested inspection frequency
- similar-asset identification
- overdue-action prioritization

But AI should **not autonomously**:

- Declare an asset safe
- Close an incident
- Approve regulatory compliance
- Approve return to service
- Downgrade critical risk
- Override a failed safety inspection

Those remain governed human decisions.

RBAC and Governance

Permissions should distinguish:

Report Hazard
Report Near Miss
Report Incident
Create Inspection
Perform Inspection
Edit Checklist
Assess Risk
Start Investigation
Create Corrective Action
Create Work Order
Change Severity
Manage Audit
Verify Action
Approve Safety
Close Incident
Close Finding
Override Risk
Return Asset to Service

For critical cases, TICVAI should support **segregation of duties** between:

Reporter → Investigator → Action Owner → Verifier → Approver

All changes must maintain:

Who | What | Before | After | When | Reason | Approval

Board 5 Acceptance Criteria

Board 5 is accepted when TICVAI can successfully manage:

Inspect → Identify Hazard/Failure → Report Hazard/Near Miss/Incident → Assess Risk → Investigate → Create Corrective Action → Generate Work Order if Required → Verify Action → Reassess Risk → Confirm Compliance → Close

across tenants, venues, assets and operational locations.

The system should maintain complete traceability between:

Asset → Inspection / Hazard / Incident → Risk → Investigation → Finding → Corrective Action → Work Order → Evidence → Verification → Residual Risk → Closure

while keeping **Safety Management and Maintenance Execution separate but fully integrated.**

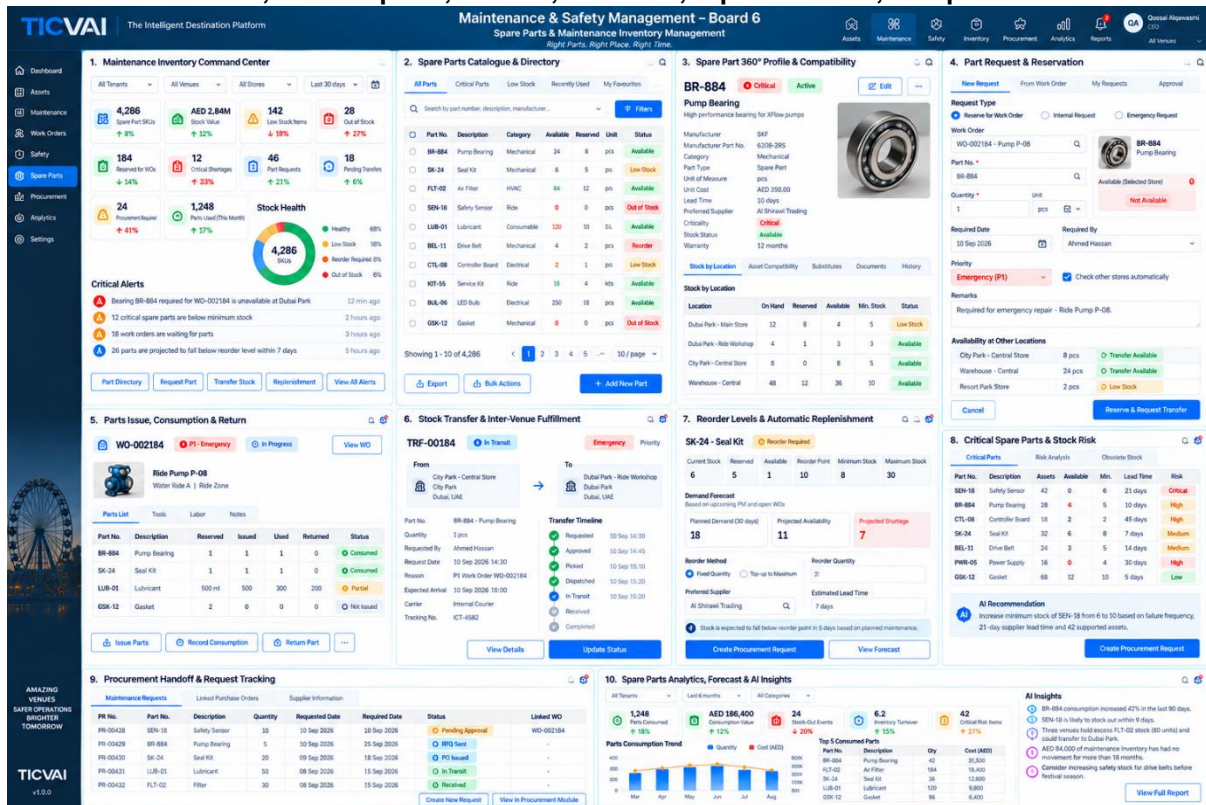
Board Boundary

Board 1 — Asset identity & lifecycle
Board 2 — Preventive maintenance planning
Board 3 — Corrective & emergency maintenance
Board 4 — Work-order execution & field operations
Board 5 — Safety, inspection & compliance
Board 6 — Spare parts & maintenance inventory
Board 7 — Procurement & maintenance resources
Board 8 — Maintenance intelligence & reporting
Board 5 = 10 screens.

Module 17 Maintenance and Safety Management

Board 6 Spare Parts and Maintenance Inventory Management

Board Objective: Provide TICVAI with a centralized spare-parts and maintenance-material management capability that connects **assets, maintenance plans, work orders, warehouses/stores, stock reservations, consumption, returns, transfers, replenishment, and procurement.**



The matrix requires TICVAI to support **spare-parts tracking, reservation and consumption, integration with inventory, and automatic procurement requests when required.**

The core principle is:

Maintenance teams must know whether the required part is available before the work begins, reserve it for the correct work order, record exactly what was consumed, and automatically trigger replenishment when stock is insufficient.

Importantly, Board 6 should **not become a duplicate of TICVAI's main Inventory & Procurement module.** It is the **maintenance-specific inventory experience**, while the central Inventory/Procurement engine remains the source of truth for stock and purchasing.

Screen 1 Maintenance Inventory Command Center

Purpose

Provide maintenance managers and store teams with an immediate overview of maintenance spare-parts availability across all venues and stores.

KPI Cards

Display:

- Total Spare Part SKUs
- Total Stock Value
- Low Stock Items
- Out-of-Stock Items
- Reserved for Work Orders
- Critical Parts Below Minimum
- Pending Part Requests
- Pending Transfers

- Procurement Required
- Parts Used This Month

Example:

KPI	Value
Spare Part SKUs	4,286
Stock Value	AED 2.84M
Low Stock	142
Out of Stock	28
WO Reservations	184
Critical Shortages	12
Part Requests	46
Pending Transfers	18

Inventory Health

Show:

- ☒ Healthy
- ☐ Low Stock
- ☒ Reorder Required
- ☒ Out of Stock
- ☐ Reserved

Stock Overview

Analyze by:

Tenant | Venue | Store | Asset Category | Part Category | Criticality

Critical Alerts

Examples:

- ☒ Bearing BR-884 required for P1 work order WO-002184 is unavailable at Dubai Park.
- ☒ 12 critical spare parts are below minimum stock.
- ☒ 18 maintenance work orders are currently waiting for parts.
- ☐ 26 parts are projected to fall below reorder level within seven days.

Quick Actions

Part Directory

Request Part

Reserve Stock

Transfer Stock

Critical Parts

Replenishment

Screen 2 Spare Parts Catalogue and Directory

Purpose

Provide a centralized searchable catalogue of all parts, consumables and maintenance materials available for asset maintenance.

Parts Table

Part No.	Description	Category	Available	Reserved	Criticality	Status
BR-884	Pump Bearing	Mechanical	24	8	Critical	Available
SK-24	Seal Kit	Mechanical	6	5	High	Low
FLT-02	Filter	HVAC	84	12	Medium	Available
SEN-18	Safety Sensor	Ride	0	0	Critical	Out of Stock

Search

Search using:

- Part number
- description
- manufacturer
- manufacturer part number
- supplier SKU
- barcode
- QR
- compatible asset
- asset model

Filters

Support:

- Tenant
- venue
- store
- category
- manufacturer
- supplier
- criticality
- stock status
- compatible asset type
- serialized/non-serialized

Part Types

Support:

Spare Part**Replacement Component****Consumable****Repair Kit****Safety Component****Electrical Component****Mechanical Component****IT and Device Component****Actions****View Part****Reserve****Request****Transfer****View Compatibility****Screen 3 Spare Part 360 Profile and Compatibility****Purpose**

Provide a complete operational record for each maintenance part and define exactly which assets it can be used with.

Header**BR 884 Pump Bearing****Critical Spare Part****Part Information**

Display:

- Internal Part ID
- manufacturer part number
- description
- category
- manufacturer
- preferred supplier
- unit of measure
- unit cost
- lead time
- criticality
- stock status

Stock by Location

Example:

Location	On Hand	Reserved	Available
Dubai Park Main Store	12	8	4
Ride Workshop	4	1	3
City Park Store	8	0	8

Asset Compatibility

Link the part to:

Asset Category**Manufacturer****Model****Individual Asset**

Example:

Pump

→ XFlow

→ P-X8

→ AST-00186

Compatibility Status**Approved****Alternative****Not Compatible****Substitute Parts**

Example:

BR-884

Alternative:

BR-884B

with:

Approved substitute — engineering approval not required.

or:

Conditional substitute — supervisor approval required.

Related Information

Tabs:

Stock | Assets | Work Orders | Suppliers | Transactions | Documents | History**Screen 4 Maintenance Part Request and Reservation****Purpose**

Allow technicians and maintenance planners to request and reserve required parts against specific maintenance work.

Request Source

A part request can originate from:

Preventive Work Order**Corrective Work Order****Emergency Work Order****Inspection****Maintenance Planner****Technician****Example****WO-002184****Ride Pump P 08**

Required:

BR 884 Bearing 1**Availability Check**

TICVAI searches:

1. Assigned venue store
2. Other stores within same venue
3. Other venue stores
4. Central warehouse

Result**Dubai Park Main Store**

Available: 0

Ride Workshop

Available: 0

City Park

Available: 8

Recommended Action**Transfer 1 unit from City Park.****Reservation**

Once approved:

Available Stock

↓

Reserved Stock

↓

Linked to:

WO-002184

This prevents another work order from consuming the same unit.

Reservation Expiry

Configure reservation expiration.

Example:

Release reservation if WO is cancelled or if reserved stock is not issued within 48 hours.

Screen 5 Parts Issue Consumption and Return**Purpose**

Control physical issue and actual consumption of maintenance stock.

Issue Workflow**Reserved**

↓

Issued to Technician

↓

Consumed

or

Returned**Example****WO-002184**

Part	Reserved	Issued	Used	Returned
BR-884 Bearing	1	1	1	0

Part	Reserved	Issued	Used	Returned
SK-24 Seal Kit	1	1	1	0
Lubricant	500ml	500ml	300ml	200ml

Store Actions

Issue

Partial Issue

Reject

Substitute

Return

Technician Confirmation

Technician confirms:

Part Received

Consumption

When technician records:

Used 1

TICVAI:

reduces inventory

-

posts part to Work Order

-

updates maintenance cost

Unused Part

If not used:

Return to Store

Stock becomes available again after required inspection/confirmation.

Damaged Part

Support:

Damaged

Defective

Scrapped

with reason and evidence.

Screen 6 Stock Transfer and Inter Venue Fulfillment

Purpose

Allow maintenance parts to be moved between stores, warehouses, venues, or maintenance workshops when local stock is unavailable.

Transfer Request

Example:

TRF-00184

From:

City Park Store

To:

Dubai Park Ride Workshop

Part:

BR-884

Quantity:

1

Reason:

P1 Work Order WO-002184

Transfer Priority

Emergency

Urgent

Normal

Workflow

Requested

↓

Approved

↓

Reserved at Source

↓

Picked

↓

Dispatched

↓

In Transit

↓

Received

↓

Available and Issued to WO

Emergency Transfer

For P1 work:

☒ **Emergency Maintenance Transfer**

TICVAI should clearly identify the operational impact.

Transfer Tracking

Capture:

- Requested by
- approved by
- source
- destination
- courier/internal runner
- dispatch time
- expected arrival
- actual arrival
- receiving user

Chain of Custody

For critical or serialized components, maintain full transfer history.

Screen 7 Reorder Levels and Automatic Replenishment

Purpose

Prevent maintenance teams from discovering critical shortages only after an asset fails.

Stock Policy

For each part/location configure:

Minimum Stock

Maximum Stock

Reorder Point

Reorder Quantity

Safety Stock

Example

BR 884 Dubai Park

Current:

4

Reserved:

3

Available:

1

Minimum:

5

Reorder Point:

6

Status

 **REPLENISHMENT REQUIRED**

Reorder Methods

Support:

Fixed Quantity

Example:

Reorder 10 units when available stock reaches 5.

Top Up to Maximum

Example:

Replenish until stock reaches 20.

Demand Aware Replenishment

TICVAI should also consider:

- Upcoming PM work orders
- Existing reservations
- historical consumption
- lead time
- asset population
- part criticality

Example

Current Available:

8

Minimum:

5

Normally:

 **Healthy**

But upcoming PM demand:

12

TICVAI should show:

● **Projected shortage: 4 units.**

This is much more useful than looking only at today's stock.

Screen 8 Critical Spare Parts and Stock Risk

Purpose

Provide special governance for components whose absence could cause significant operational or safety downtime.

Critical Spare Definition

Criticality may depend on:

- Asset criticality
- safety impact
- operational impact
- supplier lead time
- availability of substitutes
- number of assets using the part
- historical failure rate

Critical Spare Dashboard

Part	Assets Supported	Available	Minimum	Lead Time	Risk
SEN-18 Safety Sensor	42	0	6	21d	Critical
BR-884 Bearing	28	4	5	10d	High
CTRL-08 Controller	18	2	2	45d	High

Risk Example

SEN-18

42 assets depend on this part

Available:

0

Supplier Lead Time:

21 Days

Open WO:

1

Result:

● **CRITICAL STOCK RISK**

Actions

Create Procurement Request

Transfer Stock

Review Substitute

Increase Safety Stock

Notify Maintenance Manager

AI Recommendation

"Increase minimum stock of SEN-18 from 6 to 10 based on failure frequency, 21-day supplier lead time and 42 supported assets."

Screen 9 Maintenance Procurement Handoff

Purpose

Automatically hand maintenance shortages to TICVAI's procurement process without building a second procurement system.

Procurement Trigger Sources

A request may originate from:

Out of Stock Work Order

Minimum Stock Breach

Projected Shortage

Critical Spare Risk

Manual Maintenance Request

Example

PR-00428

Requested By:

Maintenance Inventory

Reason:

Critical Spare Shortage

Part:

SEN-18

Required Quantity:

10

Required Date:

20 Sep 2026

Procurement Context

Transfer:

- Part
- quantity
- preferred supplier
- alternative suppliers
- last purchase price
- expected lead time
- criticality
- related work orders
- affected assets
- required date

Handoff

Maintenance Inventory

↓

Procurement Request

↓

Central Procurement Module

Board 7 will provide the maintenance operational view of this process.

The actual purchasing lifecycle should remain within TICVAI's existing **Inventory & Procurement module**.

Status Visibility

Maintenance users should still see:

Requested**Pending Approval****RFQ****PO Issued****Supplier Confirmed****In Transit****Received**

without needing access to the complete procurement administration system.

Screen 10 Spare Parts Analytics Forecast and AI Insights

Purpose

Provide management with intelligence on maintenance consumption, shortages, cost, critical stock risk and future demand.

KPI Cards

Display:

- Parts Consumed
- Consumption Value
- Emergency Parts Usage
- Stock-Out Events
- Inventory Turnover
- Obsolete Stock
- Critical Stock Risk
- Procurement Lead Time

Consumption Analysis

Analyze by:

Tenant**Venue****Asset****Asset Category****Work Order Type****Part****Manufacturer****Supplier****Top Consumed Parts**

Example:

Part	Qty	Cost	WO Count
BR-884 Bearing	42	AED 31,500	38
FLT-02 Filter	184	AED 18,400	126
SK-24 Seal Kit	36	AED 12,600	32

Maintenance Cost Correlation

Example:

Pump P-X8 consumed AED 42,800 in spare parts during the last six months.

Forecast**Next 30 Days**

Based on planned PM:

Parts Required: 1,842

Current Available: 1,610

Projected Shortage: 232

Estimated Procurement: AED 126,000

AI Insights

Examples:

“BR-884 consumption increased 42% during the last 90 days.”

“SEN-18 is likely to stock out within nine days based on current failure rate.”

“AED 84,000 of maintenance inventory has had no movement for more than 18 months.”

“Three venues hold excess FLT-02 stock while Dubai Park is projected to experience a shortage.”

“Transferring 20 units from City Park could avoid an emergency purchase.”

Recommended Actions

Transfer Stock

Adjust Reorder Point

Create Procurement Request

Review Obsolete Stock

Review Asset Reliability

Board 6 Shared Configuration Requirements

Authorized administrators should configure:

- Spare-part categories
- part numbering
- units of measure
- manufacturers
- supplier references
- part criticality
- asset compatibility
- substitute parts
- serialized/non-serialized rules
- store/warehouse hierarchy
- maintenance stock locations
- reservation rules
- reservation expiry
- issue rules
- consumption rules
- return rules
- damaged/scrap reasons
- transfer workflows
- emergency transfer rules
- minimum stock
- maximum stock
- reorder point

- reorder quantity
- safety stock
- lead times
- replenishment rules
- procurement triggers
- approval workflows
- inventory valuation visibility
- notification rules
- stock-risk thresholds

Critical Architecture Do Not Duplicate TICVAI Inventory

This is very important for Softlabs.

We already have a broader **Inventory & Procurement module** in TICVAI.

Therefore Board 6 should **not create another inventory database** specifically for Maintenance.

The architecture should be:

TICVAI Central Inventory Engine

maintains:

SKU

Stock

Warehouse

Movement

Reservation

Cost

Supplier

Procurement

Then:

Maintenance and Safety Board 6

provides a specialized maintenance view:

Asset Compatibility

Work Order Reservation

Technician Issue

Maintenance Consumption

Critical Spare Risk

PM Demand Forecast

So if the store has:

10 × BR-884

and maintenance reserves:

4

the central inventory must immediately show:

On Hand: 10

Reserved: 4

Available: 6

There must never be one quantity in Maintenance and another quantity in Inventory.

Critical Requirement Planned vs Actual Parts

Every work order should distinguish:

Planned

What maintenance expected to use.

Reserved

What stock has been secured.

Issued

What the technician physically received.

Consumed

What was actually installed/used.

Returned

What came back to inventory.

For example:

Planned: 2

Reserved: 2

Issued: 2

Consumed: 1

Returned: 1

This is essential for inventory accuracy and maintenance costing.

Critical Requirement Asset to Part Compatibility

TICVAI should not allow technicians to blindly choose any inventory item.

The system should understand:

Asset

↓

Manufacturer

↓

Model

↓

Compatible Parts

Example:

Pump P X8

Compatible:

✓ BR-884

✓ SK-24

Alternative:

△ BR-884B

Not Compatible:

× BR-992

For emergency situations, an authorized engineer may override compatibility where policy permits, but TICVAI must capture:

Who Why Approval**Critical Requirement PM Demand Forecasting**

Board 6 should consume upcoming maintenance demand from Board 2.

Example:

Next Month

100 turnstiles require PM.

Each requires:

1 × Filter F-08

Inventory:

70

Open Purchase Order:

10

Projected Availability:

80

Required:

100

TICVAI should identify:

Projected shortage: 20 filters.

This should happen **before the 100 work orders reach execution.**

That is one of the most valuable integrations between Boards 2 and 6.

Relationship With Board 4 Work Orders

The execution flow should be:

Work Order

↓

Required Parts

↓

Check Availability

↓

Reserve

↓

Issue

↓

Technician Receives

↓

Consume and Return

↓

Update Inventory

↓

Post Actual Cost to Work Order

If unavailable:

Part Request

↓

Transfer

or

Procurement

The work order can automatically change to:

WAITING PARTS

when required stock is unavailable.

Once stock becomes available:

“Required parts for WO-002184 are now available.”

The maintenance team can resume the work.

Relationship With Board 2 Preventive Maintenance

Preventive Maintenance plans can define:

Standard Parts Requirement

Example:

HVAC Quarterly Service

Expected:

- Filter × 2
- Belt × 1
- Lubricant × 250ml

When upcoming PM is scheduled, TICVAI can calculate total demand automatically.

This enables:

Maintenance Schedule

→ **Parts Forecast**

→ **Reservation**

→ **Replenishment**

before technicians arrive.

Relationship With Board 3 Emergency Maintenance

Emergency work should have special stock handling.

Example:

P1 Emergency

Required part unavailable locally.

TICVAI searches:

Same Venue

↓

Nearby Store

↓

Other Venue

↓

Central Warehouse

↓

Approved Substitute

↓

Emergency Procurement

and recommends the fastest governed fulfillment route.

Serialized Critical Components

Certain components should be individually traceable.

Examples:

- Controllers
- motors
- safety sensors
- high-value electronics
- critical ride components

For these:

Part SKU

-

Serial Number

must be recorded.

When installed:

Component Serial SN 884221

moves from:

Warehouse

↓

WO-002184

↓

Installed on Asset AST 00186

If removed later, TICVAI knows exactly:

when it was installed

where it was installed

which technician installed it

which work order installed it

when it was removed

This creates a powerful component lifecycle history.

QR and Barcode Operations

The field and store experience should support scanning.

Storekeeper:

Scan WO

↓

Scan Part

↓

Issue

Technician:

Scan Part

↓

Confirm Received

Technician after repair:

Scan Installed Component

↓

Record Consumption

This reduces manual part-number entry and stock errors.

AI Governance

AI may assist with:

- Demand forecasting
- stock-out prediction
- safety-stock recommendations
- excess inventory detection
- inter-venue transfer recommendations
- substitute suggestions
- abnormal consumption detection
- supplier lead-time patterns
- part-to-asset reliability correlation

However, AI should not automatically:

- approve purchases,
- substitute safety-critical components,
- change approved engineering compatibility,
- scrap high-value inventory,
- or override stock governance

without the configured authorization process.

RBAC and Governance

Permissions should distinguish:

View Stock

View Cost

Request Part

Reserve Part

Issue Part

Consume Part

Return Part

Transfer Part

Approve Transfer

Manage Compatibility

Approve Substitute

Change Reorder Point

Change Safety Stock

Create Procurement Request

Scrap Part

Manage Critical Spare Classification

Financial information such as:

Unit Cost

Stock Value

Supplier Price

may be hidden from technicians while remaining available to authorized managers.

All stock movements must record:

Part | Quantity | From | To | User | Date/Time | Source Record | Reason

Board 6 Acceptance Criteria

Board 6 is accepted when TICVAI can successfully manage:

Identify Part → Check Compatibility → Check Availability → Request → Reserve → Issue → Consume/Return → Transfer if Required → Replenish → Trigger Procurement → Forecast Future Demand → Analyze

maintenance spare parts across tenants and venues.

The system must preserve the complete chain:

Asset → Maintenance Requirement → Work Order → Required Part → Reservation → Issue → Consumption → Inventory Movement → Cost → Replenishment / Procurement

while maintaining **one central TICVAI Inventory source of truth**.

Board Boundary

Board 1 — Asset identity & lifecycle

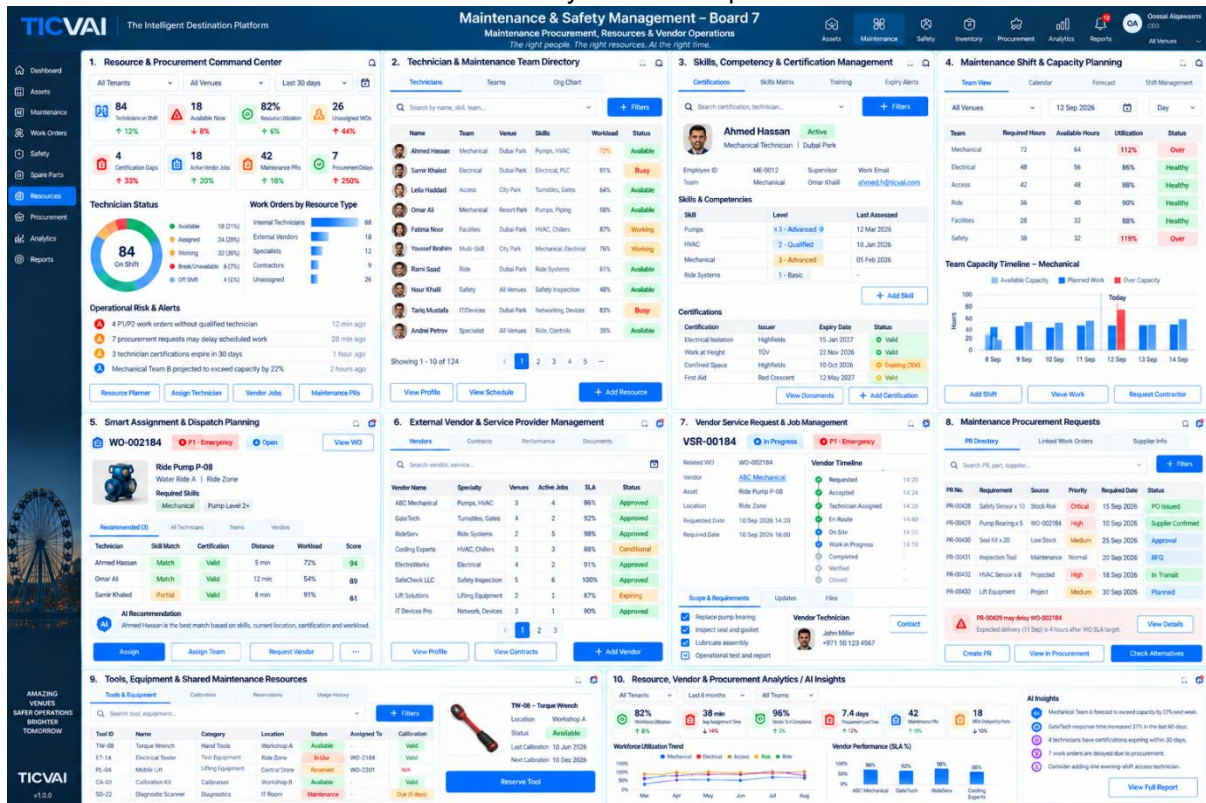
Board 2 — Preventive maintenance planning

Board 3 — Corrective & emergency maintenance

Board 4 — Work-order execution & field operations
Board 5 — Safety, inspection & compliance
Board 6 — Spare parts & maintenance inventory
Board 7 — Maintenance procurement & resource operations
Board 8 — Maintenance intelligence & reporting
Board 6 = 10 screens.

Board 7 Maintenance Procurement Resources and Vendor Operations

Board Objective: Provide TICVAI with the operational layer for coordinating the **people, external vendors, procurement dependencies, tools, certifications, workload, shifts, and service providers** required to execute maintenance efficiently across multiple tenants and venues.



Board 7 should connect maintenance operations with TICVAI's existing **Procurement, Inventory, HR/resource information, Finance, and Work Order engines** rather than duplicating those systems.

The core principle is:

A maintenance job should not only tell TICVAI what needs to be done; the platform should also know who can do it, whether they are available and qualified, what external support is required, and whether procurement dependencies could delay the work.

Screen 1 Maintenance Resource and Procurement Command Center

Purpose

Provide maintenance leadership with one operational overview of workforce capacity, vendor activity, procurement dependencies, and resource risks.

KPI Cards

Display:

- Active Technicians
- Available Technicians
- Technicians on Shift
- Resource Utilization
- Unassigned Work Orders
- Skill/Certification Gaps
- Active Vendors
- Vendor Jobs in Progress
- Maintenance PRs

- Procurement Delays
- SLA at Risk Due to Resources
- SLA at Risk Due to Parts

Example:

KPI	Value
Technicians on Shift	84
Available Now	18
Utilization	82%
Unassigned WOs	26
Certification Gaps	4
Active Vendor Jobs	18
Maintenance PRs	42
Procurement Delays	7

Operational Risk Panel

Examples:

- Four P1/P2 work orders currently have no qualified technician available.
- Seven maintenance procurement requests may affect scheduled work.
- △ Three technician certifications expire within 30 days.
- Mechanical Team B is projected to exceed capacity by 22% tomorrow.

Breakdown

Analyze by:

Tenant | Venue | Team | Skill | Vendor | Work Type | Priority

Quick Actions

Resource Planner

Assign Technician

Vendor Jobs

Maintenance Procurement

Certification Gaps

Capacity Forecast

Screen 2 Technician and Maintenance Team Directory

Purpose

Maintain the operational maintenance-resource directory used for work assignment and capacity planning.

Technician Directory

Technician	Team	Venue	Skills	Shift	Workload	Status
Ahmed Hassan	Mechanical	Dubai Park	Pumps/HVAC	08–17	72%	Available
Samir Khaled	Electrical	Dubai Park	Electrical/Controls	08–17	91%	Busy
Leila Haddad	Access	City Park	Gates/Turnstiles	14–23	64%	Available

Technician Profile

Capture/display:

- Employee/resource ID
- name
- primary venue
- maintenance team
- supervisor
- skills
- certifications
- authorization level
- working hours
- shift
- availability
- current workload

Resource Types

Support:

Internal Technician

Maintenance Supervisor

Engineer

Specialist

Safety Inspector

Contractor

External Vendor Technician

Status



Available



Assigned



Working



Break / Unavailable



Off Shift



Restricted / Certification Issue

Actions

View Schedule

View Skills

Assign Work

View Workload

View Certifications

Screen 3 Skills Competency and Certification Management

Purpose

Ensure maintenance work is assigned only to personnel with the required skills, qualifications, or certifications.

Skill Library

Examples:

- Mechanical
- Electrical
- HVAC
- Pumps

- PLC/Controls
- Access Control
- Turnstiles
- Ride Maintenance
- Fire Systems
- Networking
- Device Diagnostics
- Welding
- Safety Inspection

Skill Level

Configure:

Level 1 Basic

Level 2 Qualified

Level 3 Advanced

Level 4 Specialist

Certification Record

Capture:

- Certification
- issuing authority
- issue date
- expiry date
- certificate/document
- verification status
- applicable work types

Example

Ahmed Hassan

Pump Maintenance

Level:

3 — Advanced

Electrical Isolation

Certification:

Valid until 15 Jan 2027

Expiry Alert

⚠ Electrical Isolation certification expires in 30 days.

Assignment Control

If a WO requires:

Electrical Isolation Certification

and technician certification is expired:

Assignment Blocked

or, depending on tenant policy:

Supervisor Override Required

Certification Dashboard

Show:

Valid | Expiring Soon | Expired | Missing

Screen 4 Maintenance Shift and Capacity Planning

Purpose

Plan maintenance workforce capacity against upcoming preventive, corrective, safety and operational work.

Resource Calendar

Views:

Day

Week

Month

Team

Technician

Capacity Example

Mechanical Team 12 September

Available:

64 hours

Planned Work:

72 hours

Utilization:

112

 **Over Capacity**

Demand Sources

Capacity planning should include:

- Scheduled PM
- Open corrective WOs
- Emergency work
- safety inspections
- planned shutdown work
- project/installations
- expected maintenance demand

Capacity by Team

Team	Required	Available	Utilization	Status
Mechanical	72h	64h	112%	Over
Electrical	48h	56h	86%	Healthy
Access	42h	48h	88%	Healthy
Safety	38h	32h	119%	Over

Actions

Move Work

Reassign Technician

Add Shift

Request Contractor

Use External Vendor

Forecast

Show capacity:

Next 7 Days**Next 30 Days**

This expands the resource-planning view introduced in Board 2 without creating a second scheduling engine.

Screen 5 Smart Assignment and Dispatch Planning**Purpose**

Help supervisors assign work based on operational suitability rather than simply selecting a technician name.

Work Order

WO-002184

P1 Emergency**Pump P 08**

Required:

Mechanical and Pump Level 2**Candidate Ranking**

Technician	Skill	Certification	Distance	Workload	Score
Ahmed	Match	Valid	5 min	72%	94
Omar	Match	Valid	12 min	54%	89
Samir	Partial	Valid	8 min	91%	61

Assignment Factors

TICVAI should consider:

Skill

-

Certification

-

Availability

-

Shift

-

Venue and Location

-

Workload

-

Priority

-

Asset familiarity**AI Recommendation**

“Ahmed Hassan is recommended because he holds the required pump certification, is currently in Ride Zone, and has lower response time.”

Conflict Detection

Example:

⚠ Ahmed is already assigned to a P2 job scheduled to start in 20 minutes.

Actions**Assign****Assign Team****Override Recommendation****Request Vendor****Governance**

AI recommends.

The authorized supervisor remains responsible for assignment.

Screen 6 External Vendor and Service Provider Management**Purpose**

Manage external maintenance companies and specialists used when work cannot or should not be performed internally.

Vendor Directory

Vendor	Specialty	Venues	Active Jobs	SLA	Status
ABC Mechanical	Pumps/HVAC	3	4	96%	Approved
GateTech	Turnstiles	4	2	92%	Approved
RideServ	Ride Systems	2	5	98%	Approved

Vendor Profile

Capture/display:

- Vendor
- service categories
- approved asset categories
- contact
- contract
- service coverage
- supported venues
- operating hours
- emergency availability
- response SLA
- certifications
- insurance documents
- contract expiry

Vendor Status

- ☒ Approved
- ☐ Conditional
- ☒ Expiring
- ☒ Suspended
- ☐ Inactive

Service Capabilities

Example:

GateTech

Approved for:

✓ Turnstile Mechanical

✓ Gate Controller

✓ Reader Installation

Not approved:

✗ Ride Equipment

Vendor Documents

Maintain:

- Contract
- insurance
- certifications
- permits
- SLA agreement
- rate card
- supporting documentation

The commercial contract can remain linked to the appropriate central TICVAI procurement/contract record.

Screen 7 Vendor Maintenance Job and Service Request

Purpose

Manage maintenance activities that must be performed by an external provider.

Vendor Service Request

Example:

VSR-00184

Related:

WO-002184

Vendor:

ABC Mechanical

Asset:

Pump P 08

Priority:

P1

Scope

Transfer from WO:

- Asset
- location
- fault
- work required
- priority
- required date/time
- safety requirements
- attachments

Vendor Workflow

Requested

↓

Accepted

↓

Technician Assigned

↓

En Route

↓

On Site

↓

Work in Progress

↓

Vendor Completed

↓

TICVAI Verification

↓

Closed**Vendor ETA**

Example:

Requested: 14:20**Accepted:** 14:24**ETA:** 14:55**SLA**

Track:

Acceptance Time**Response Time****Arrival Time****Completion Time****Vendor Evidence**

Allow vendor or authorized internal user to provide:

- service report
- photos
- technician notes
- parts used
- test results
- invoice reference

Final VerificationVendor marking work complete should **not automatically close the TICVAI Work Order**.

Internal verification may still be required.

Screen 8 Maintenance Procurement Request and Dependency Tracking**Purpose**

Provide maintenance teams with visibility over procurement requests affecting maintenance execution.

This is the maintenance-facing layer of TICVAI's existing procurement engine.

Maintenance PR Directory

PR	Requirement	Source	Priority	Required	Status
PR-00428	Safety Sensor ×10	Stock Risk	Critical	15 Sep	PO Issued
PR-00429	Pump Bearing ×5	WO	P1	10 Sep	Supplier Confirmed

PR	Requirement	Source	Priority	Required	Status
PR-00430	Specialist Service	WO	P2	12 Sep	Approval
PR-00431	Inspection Tool	Maintenance	Normal	20 Sep	RFQ

Request Types

Support maintenance requests for:

Spare Parts

Consumables

Tools

Equipment

External Services

Specialist Services

Rental Equipment

Procurement Visibility

Maintenance users can see:

Requested

↓

Approval

↓

RFQ

↓

Supplier Selection

↓

PO

↓

Supplier Confirmation

↓

Delivery

↓

Received

Dependency

Show affected work.

Example:

PR-00429

Linked to:

WO-002184

Current WO Status:

WAITING PART

Expected Delivery:

11 Sep — 10:00

Risk

● Expected delivery is 4 hours after the WO SLA target.

Actions**Escalate Procurement****Check Alternative Stock****Review Substitute****Request Emergency Purchase**

subject to permissions.

Screen 9 Tools Equipment and Shared Maintenance Resources**Purpose**

Manage maintenance resources that are not consumed like spare parts but must be available to perform work.

Resource Types

Examples:

- Diagnostic Tools
- Calibration Equipment
- Lifting Equipment
- Torque Wrenches
- Electrical Testers
- Mobile Work Platforms
- Specialized Ride Tools
- Safety Equipment
- Maintenance Vehicles
- Rental Equipment

Resource Directory

Resource	Location	Status	Assigned To	Calibration
Torque Wrench TW-08	Workshop A	Available	—	Valid
Tester ET-14	Ride Zone	In Use	WO-2184	Valid
Lift PL-04	Central Store	Reserved	WO-2201	N/A

Resource Status**Available****Reserved****Issued****In Use****Maintenance****Calibration Due****Unavailable****Reservation**

Work Order can require:

1 Electrical Tester**1 Torque Wrench**

TICVAI reserves them for the planned maintenance window.

Calibration Control

For tools requiring calibration:



Tool calibration expired.

TICVAI can prevent use on governed work types.

Check Out and Check In

Support:

Scan Tool

↓

Issue

↓

Use

↓

Return

Lost and Damaged

Capture:

Lost

Damaged

Calibration Failure

and create follow-up actions where required.

Screen 10 Resource Vendor and Procurement Analytics and AI Insights

Purpose

Provide management with consolidated intelligence on maintenance workforce performance, vendor performance, resource utilization, procurement delays, and operational bottlenecks.

KPI Cards

Display:

- Workforce Utilization
- Average Assignment Time
- Technician Productivity
- Vendor SLA Compliance
- Vendor Response Time
- Procurement Lead Time
- WOs Delayed by Parts
- WOs Delayed by Resources
- Overtime Hours
- External Maintenance Cost

Workforce Analysis

Analyze:

Work Orders per Technician

Average Repair Time

Utilization

Response Time

First Time Fix Rate

Vendor Performance

Example:

Vendor	Jobs	SLA	Avg Response	Rework
ABC Mechanical	42	96%	38m	2%

Vendor	Jobs	SLA	Avg Response	Rework
GateTech	28	91%	54m	7%
RideServ	18	98%	26m	1%

Procurement Analysis

Show:

- Average PR-to-PO
- supplier lead time
- emergency procurement
- delayed maintenance due to purchasing
- spend by maintenance category

Resource Bottlenecks

Example:

Electrical Team utilization: 118%

Ride-certified technicians available: 2

Required next week: 4

AI Insights

Examples:

“Mechanical Team B is forecast to exceed capacity by 22% next week.”

“Four planned PM jobs require ride-certified technicians, but only two qualified technicians are scheduled.”

“GateTech’s average response time has increased 31% during the last 60 days.”

“Seven work orders experienced procurement-related delays this month.”

“Adding one evening-shift access-control technician could reduce projected assignment delays by approximately 18%.”

AI Recommendations

Rebalance Workload

Adjust Shift

Assign Alternative Team

Use Approved Vendor

Escalate Procurement

Review Vendor Performance

Plan Certification and Training

Board 7 Shared Configuration Requirements

Authorized administrators should configure:

- Maintenance teams
- resource types
- technician roles
- skills
- competency levels
- certification types
- certification validity
- skill requirements by work type
- assignment restrictions
- technician availability

- shifts
- working calendars
- capacity rules
- utilization thresholds
- overtime rules
- smart-assignment factors
- vendor categories
- approved vendor capabilities
- vendor SLA
- vendor service areas
- vendor status
- vendor documentation requirements
- service-request workflows
- procurement request types
- maintenance procurement priorities
- procurement escalation rules
- shared tool categories
- tool reservation rules
- calibration requirements
- check-out/check-in rules
- approval workflows
- notification rules

Critical Architecture Do Not Build Another HR System

Board 7 should manage the **operational maintenance profile** of a technician.

It should not become TICVAI's HR/payroll system.

The maintenance profile needs:

Who is available?

What can they do?

What are they certified to do?

Where are they?

What shift are they working?

What work is already assigned?

Employee master information can come from:

HR Integration / User Management / External Workforce System

Therefore:

Employee and User

↓

Maintenance Resource Profile

↓

Skills Certifications Availability Workload

↓

Work Order Assignment

Critical Architecture Do Not Duplicate Procurement

Similar to Board 6, TICVAI already has an **Inventory & Procurement module**.

Board 7 should therefore provide:

Maintenance Procurement Operations

while the central procurement engine owns:

Supplier → RFQ → Quotation → Approval → PO → Receipt → Finance

Maintenance owns the context:

Why do we need it?

Which asset requires it?

Which WO depends on it?

How urgent is it?

What happens if it is late?

This separation is important.

Critical Requirement Maintenance Dependency Engine

I recommend adding a dependency concept to work orders.

Example:

WO-002184

Cannot proceed because:

Part BR 884

WAITING

and:

Vendor Specialist

CONFIRMED 15:00

and:

Torque Wrench

AVAILABLE

TICVAI should show:

Work Readiness

2 and 3 Dependencies Ready

● NOT READY

When the part arrives:

3 / 3 READY

● READY FOR EXECUTION

This would significantly improve maintenance planning.

Critical Requirement Resource Qualification

The system should differentiate:

Skill

Technician knows how to perform the work.

from:

Certification

Technician is formally authorized to perform the work.

For example:

Ahmed may have:

Electrical Skill Level 4

but:

High Voltage Certification Expired

For a governed high-voltage WO:

Assignment must be blocked.

This distinction is important for safety and compliance.

Critical Requirement External Vendor Integration

External vendors should not require unrestricted access to the TICVAI tenant.

I recommend a controlled:

Vendor Service Portal and Restricted Vendor Workspace

where the vendor sees only:

- Assigned service requests
- relevant asset information
- location
- work scope
- required documentation
- permitted attachments
- status actions

The vendor should **not see unrelated assets, financial information, other work orders, customer information, or other vendors**.

Relationship With Board 4 Work Order Engine

Board 4 remains the source of truth for maintenance execution.

Board 7 supplies:

Technician

Team

Vendor

Tool

Procurement Dependency

to the Work Order.

Architecture:

WO Requirement

↓

Board 7 Resource Matching

↓

Technician and Vendor and Tool

↓

Board 4 Execution

↓

Actual Resource Usage

↓

Board 7 Analytics

There should be no separate vendor work order replacing the central WO.

Relationship With Board 6 Spare Parts

Board 6 answers:

Do we have the part?

Board 7 answers:

If we do not have it, where is the procurement process and when will it arrive?

Example:

WO

↓

Part Required

↓

Board 6:

Out of Stock

↓

Board 7:

Procurement PR-00429

↓

Central Procurement:

PO Issued

↓

Supplier:

Delivery ETA

↓

Board 7:

Dependency ETA

↓

Board 4:

WAITING PART

This creates end-to-end visibility without duplication.

Relationship With Safety

Certification and safety requirements from Board 5 should influence assignment.

Example:

Work at Height WO

requires:

Work at Height Certification

and:

Valid Safety Permit

Board 7 validates the technician qualification.

Board 5 validates the safety requirement.

Board 4 controls execution.

Therefore:

Qualified Person Valid Safety Conditions Available Resources

↓

Work Can Start**Resource Costing**

Board 7 should provide actual resource information to maintenance costing.

For each work order:

Internal Labor

•

Overtime

•

Vendor Service

-

Rental Equipment

-

Parts from Board 6

=

Total Maintenance Cost

Financial posting and accounting should still flow to TICVAI Finance rather than being maintained as an independent maintenance ledger.

AI Governance

AI may assist with:

- technician recommendations
- workload balancing
- shift forecasting
- skill-gap detection
- certification planning
- vendor recommendations
- vendor-performance patterns
- procurement-delay prediction
- resource bottleneck forecasting
- maintenance-demand forecasting

AI must not automatically:

- authorize unqualified technicians,
- bypass expired certifications,
- approve procurement,
- select a vendor where formal procurement approval is required,
- override safety requirements,
- or approve overtime outside configured governance.

RBAC and Governance

Permissions should distinguish:

View Resources**Manage Maintenance Profile****Manage Skills****Manage Certifications****Assign Technician****Override Assignment****Manage Shift****Request Vendor****Approve Vendor Assignment****Manage Vendor Record****Create Maintenance PR****Escalate Procurement****Reserve Tool****Issue Tool****Manage Calibration****View Labor Cost****View Vendor Cost****View Procurement Cost**

Sensitive commercial information should be permission-controlled.

All critical changes must maintain:

Who | What | Before | After | When | Reason | Approval**Board 7 Acceptance Criteria**

Board 7 is accepted when TICVAI can successfully:

Identify Work Requirement → Determine Required Skill/Certification → Check Capacity → Match Technician/Team/Vendor → Confirm Tools → Track Procurement Dependencies → Dispatch Resources → Monitor Vendor/Procurement SLA → Feed Resources into Work Order → Analyze Performance

across tenants and venues.

The system must preserve the operational chain:

Work Order → Required Skills/Resources → Qualified Technician / Vendor → Required Tool → Required Part/Procurement → Readiness → Execution → Actual Resource Usage → Cost → Performance

while maintaining **one central Work Order Engine, one Inventory/Procurement source of truth, and no duplicate HR system.**

Board Boundary**Board 1 — Asset identity & lifecycle****Board 2 — Preventive maintenance planning****Board 3 — Corrective & emergency maintenance****Board 4 — Work-order execution & field operations****Board 5 — Safety, inspection & compliance****Board 6 — Spare parts & maintenance inventory****Board 7 — Maintenance procurement, resources & vendor operations****Board 8 — Maintenance intelligence & reporting****Board 7 = 10 screens.**

Board 8 Maintenance Intelligence Analytics and Executive Reporting

Board Objective: Provide TICVAI with the consolidated intelligence and reporting layer for the complete Maintenance & Safety Management module, giving operational managers, maintenance leadership, venue management, and executives visibility into **asset reliability, preventive and corrective maintenance, downtime, work-order performance, safety, SLA compliance, maintenance costs, resources, vendors, spare parts, and future risks.**



Board 8 is not another operational transaction board. It consolidates the data created across **Boards 1–7** and converts it into actionable operational intelligence.

The core principle is:

TICVAI should not only tell management what happened. It should explain why it happened, what is likely to happen next, and where management attention is required.

Screen 1 Maintenance Executive Command Center

Purpose

Provide senior management with a consolidated overview of maintenance and safety performance across the entire organization.

Global Filters

Support:

Tenant | Venue | Region | Asset Category | Asset | Date Range

Executive KPI Cards

Display:

- Total Assets
- Asset Availability %
- Assets Out of Service
- Maintenance Compliance %
- Open Work Orders

- Critical Work Orders
- Total Downtime
- MTTR
- MTBF
- Safety Compliance %
- Maintenance Cost
- SLA Compliance %

Example:

KPI	Value
Assets	8,426
Availability	98.4%
Out of Service	42
PM Compliance	96.8%
Open WOs	286
Critical WOs	12
SLA Compliance	95.7%
Safety Compliance	97.2%

Executive Health Indicators

Show:

- ☒ Healthy
- ☐ Attention
- ☐ At Risk
- ☐ Critical

Venue Performance

Example:

Venue	Availability	PM	Safety	SLA	Status
Dubai Park	98.8%	97%	98%	96%	Healthy
City Park	97.2%	94%	96%	91%	Attention
Resort Park	99.1%	98%	99%	98%	Healthy

Executive Alerts

Examples:

- ☒ Ride Zone A experienced 18% more downtime this month.
- ☒ Mechanical Team B is projected to exceed capacity next week.
- ☐ 12 critical spare parts are below required safety stock.
- ☐ Preventive-maintenance compliance improved by 2.4%.

Quick Navigation**Assets****Reliability****Maintenance****Safety****Costs****SLA****AI Insights****Screen 2 Asset Performance and Reliability Analytics****Purpose**

Measure how reliably assets perform and identify assets, models, manufacturers, or locations generating excessive maintenance.

Reliability KPIs

Display:

- Asset Availability
- MTBF
- MTTR
- Failure Rate
- Repeat Failure Rate
- Planned vs Unplanned Maintenance
- Asset Downtime
- Reliability Trend

MTBF**Mean Time Between Failures**

Example:

Pump P 08

MTBF = 286 hours

MTTR**Mean Time to Repair**

Example:

MTTR = 2h 18m

Reliability Ranking

Asset	Availability	MTBF	MTTR	Failures	Risk
Pump P-08	91.8%	286h	2.3h	8	High
Gate-04	96.2%	624h	0.8h	4	Medium
AHU-12	98.4%	1,240h	1.2h	2	Low

Analysis Dimensions

Allow comparison by:

Asset

Asset Type

Asset Category

Manufacturer

Model

Venue

Location

Age

Criticality

Bad Actor Identification

TICVAI should identify:

Top Failing Assets

and:

Top Failing Asset Models

This is important because one problematic model may be installed across several venues.

Reliability Trend

Show:

Last 12 Months

with:

Availability | Failure Rate | MTBF | MTTR

Screen 3 Downtime and Operational Impact Analytics

Purpose

Measure how maintenance issues affect actual venue and asset operations.

Downtime KPIs

Display:

- Total Downtime
- Planned Downtime
- Unplanned Downtime
- Critical Asset Downtime
- Average Downtime Event
- Operational Availability
- Guest-Affecting Downtime
- Revenue-Affecting Downtime

Downtime Classification

Separate:

Planned

Example:

Preventive maintenance.

Unplanned

Example:

Unexpected pump failure.

Operational

Asset technically available but intentionally unavailable due to operations.

Safety Restriction

Asset unavailable because safety approval is pending.

Example**Water Ride A**

Total Downtime:

18h 42m

Planned:

4h 10m

Unplanned:

14h 32m

Impact Analysis

Link downtime to:

- Attraction
- admission lane
- retail point
- F&B
- venue area
- operational capacity

Timeline

Show downtime periods visually.

Example:

08:00 ————— 12:20

Root Cause

Break downtime by:

Mechanical**Electrical****Device****Parts****Vendor****Safety****Operations****Important Metric**

TICVAI should distinguish:

Technical Availability

from:

Operational Availability

An asset may be connected and technically healthy but unavailable for guest operation.

Screen 4 Maintenance Performance and Work Order Analytics**Purpose**

Measure the performance of the maintenance organization and work-order lifecycle.

Work Order KPIs

Display:

- WOs Created
- WOs Completed
- WOs Open
- Overdue WOs

- First-Time Fix Rate
- Average Response Time
- Average Repair Time
- Average Closure Time
- Reopened WOs
- Backlog

Work Order Funnel

Example:

Created

1,842

↓

Assigned

1,794

↓

Started

1,728

↓

Completed

1,684

↓

Verified

1,642

↓

Closed

1,618

Time Breakdown

Measure separately:

Assignment Time

Acknowledgement Time

Travel Time

Repair Time

Waiting Parts

Waiting Vendor

Verification Time

Example

Average WO Duration

5h 24m

Breakdown:

Assignment: 12m

Travel: 18m

Repair: 2h 04m

Waiting: 2h 36m

Verification: 14m

This allows management to see that the biggest problem may not be technician repair time.

Backlog Aging

Show:

0 1 Day**2 3 Days****4 7 Days****8 30 Days****30 Days****Screen 5 Preventive vs Corrective Maintenance Intelligence****Purpose**

Determine whether the maintenance strategy is preventing failures effectively.

KPIs

Display:

- PM Compliance
- PM Completed On Time
- PM Overdue
- Corrective WOs
- Emergency WOs
- Preventive/Corrective Ratio
- Failures After PM
- PM Effectiveness

Comparison

Example:

Metric	Preventive	Corrective
Work Orders	1,284	558
Labor Hours	2,840	2,160
Cost	AED 286K	AED 412K
Downtime	84h	286h

Failure After Maintenance

TICVAI should identify:

Asset failed within X days after PM completion.

Example:

Pump P 08

PM Completed:

01 Sep

Failure:

06 Sep

5 Days After PM

This may indicate:

- Checklist issue
- PM frequency issue
- poor maintenance quality
- underlying reliability issue

PM Effectiveness

Analyze:

Before PM strategy change

versus:

After PM strategy change**AI Recommendation**

“Reducing P-X8 bearing inspection from 90 to 75 days may reduce unplanned failures.”

Screen 6 Safety Compliance and Incident Analytics**Purpose**

Provide consolidated reporting for Board 5 safety operations.

Safety KPIs

Display:

- Safety Compliance %
- Inspections Completed
- Inspection Pass Rate
- Open Hazards
- Near Misses
- Safety Incidents
- Critical Incidents
- Overdue Actions
- Repeat Findings
- Average Closure Time

Safety Trend

Show:

Incidents**Hazards****Near Misses****Failed Inspections**

over time.

Risk Heatmap

Display:

Likelihood Consequence

with active risk count.

Safety Analysis

Filter by:

Venue**Location****Asset****Asset Category****Incident Type****Severity****Inspection Type****Department****Repeat Safety Findings**

Example:

Emergency access obstruction — 8 occurrences.

Gate safety sensor — 6 occurrences.

Missing inspection evidence — 5 occurrences.

Leading vs Lagging Indicators

I recommend TICVAI distinguish:

Leading Indicators

- inspections completed
- hazards reported
- near misses reported
- corrective actions completed
- training/certification status

from:

Lagging Indicators

- incidents
- injuries
- failures
- downtime

This provides a stronger safety-management picture.

Screen 7 Maintenance Cost and Lifecycle Economics

Purpose

Understand how much assets actually cost to maintain and identify when continued repair becomes economically inefficient.

Cost KPIs

Display:

- Total Maintenance Cost
- Labor Cost
- Parts Cost
- Vendor Cost
- Emergency Maintenance Cost
- Preventive Maintenance Cost
- Cost per Asset
- Cost per Operating Hour

Cost Composition

Example:

AED 4.82M

Labor: 32%

Parts: 38%

Vendors: 22%

Other: 8%

Asset Cost

Example:

Pump P 08

Purchase Cost:

AED 28,000

Maintenance Cost — 24 months:

AED 18,400

Downtime Cost/Impact:

AED 12,000

Total Cost of Ownership Context

TICVAI can compare:

Acquisition

-

Maintenance

-

Parts

-

External Services

-

Downtime

Repair vs Replace

Example:

Asset has consumed 66% of original acquisition value in maintenance during the last 24 months.

AI Recommendation

“Consider replacement review for Pump P-08 based on rising repair cost, declining MTBF and increasing downtime.”

This should create a recommendation, not automatically retire the asset.

Screen 8 SLA Workforce Vendor and Parts Performance

Purpose

Consolidate operational dependencies affecting maintenance performance.

SLA KPIs

Display:

- Response SLA
- Resolution SLA
- PM SLA
- Vendor SLA
- Procurement SLA
- Safety Action SLA

SLA Breakdown

Analyze by:

Venue**Priority****Team****Technician****Vendor****Work Type****Workforce Performance**

Show:

- Utilization
- workload
- response time
- completion rate
- first-time fix rate
- overtime
- skill coverage

Vendor Performance

Example:

Vendor	Jobs	SLA	Response	Rework
ABC Mechanical	42	96%	38m	2%
GateTech	28	91%	54m	7%
RideServ	18	98%	26m	1%

Parts Performance

Show:

- Stock-out events
- WO delays due to parts
- emergency purchases
- critical shortages
- inventory turnover

Bottleneck Analysis

Example:

WO Delay Causes

Parts — 38%

Vendor — 24%

Resource — 18%

Approval — 12%

Other — 8%

This helps management address the actual source of maintenance delay.

Screen 9 Reporting Center and Scheduled Reports**Purpose**

Provide configurable operational, compliance and executive reporting without requiring every user to build analytics manually.

Report Library

Examples:

Asset Register Report

Asset Reliability Report

Preventive Maintenance Compliance

Corrective Maintenance Report

Work Order Performance

Downtime Report

Safety Incident Report

Safety Compliance Report

Maintenance Cost Report

Spare Parts Consumption

Vendor Performance

SLA Performance

Report Builder

Allow users to select:

Data Source

Fields

Filters

Grouping

Sorting

Date Range

Output

Support:

On Screen

PDF

Excel

CSV

where appropriate.

Scheduled Reports

Example:

Weekly Maintenance Performance

Recipients:

Maintenance Manager

Venue GM

Schedule:

Every Monday — 08:00

Distribution

Subject to permissions:

Email

Dashboard Inbox

Download

API

Multi Tenant Governance

Reports must always respect:

Tenant**Venue****User Permission****Data Visibility**

A venue manager must not accidentally receive another client's maintenance information.

Screen 10 AI Maintenance Intelligence and Natural Language Analytics

Purpose

Provide TICVAI's AI layer across the entire Maintenance & Safety Management module.

This should connect with TICVAI's broader **Reporting / Power BI / AI Analytics strategy**, rather than creating an isolated maintenance AI engine.

AI Query

Allow authorized users to ask questions such as:

"Which assets caused the most downtime this month?"

"Why did Dubai Park maintenance SLA decrease?"

"Show pumps with more than three corrective failures in 90 days."

"Which spare parts are likely to stock out next month?"

"Which assets should we consider replacing?"

"Which vendors have the highest rework rate?"

"What are our main safety risks?"

AI Response

TICVAI should provide:

Answer

"Dubai Park SLA decreased from 96.4% to 92.1%, primarily due to seven delayed corrective work orders."

Drivers

Parts Delay: 4 WOs

Vendor Delay: 2 WOs

Resource Delay: 1 WO

Evidence

Allow users to drill into the actual:

Work Orders**Assets****Parts****Vendor Jobs****Recommended Actions**

Example:

Increase BR 884 Safety Stock

Review GateTech SLA

Add Mechanical Capacity

Predictive Insights

Where sufficient data exists:

Failure Risk

Stock Out Risk

SLA Breach Risk

Capacity Risk

Safety Risk

Replacement Risk

Board 8 Shared Configuration Requirements

Authorized administrators should configure:

- KPI definitions
- KPI formulas
- MTBF rules
- MTTR rules
- availability formulas
- downtime classifications
- planned/unplanned downtime
- maintenance compliance formulas
- PM effectiveness windows
- repeat-failure thresholds
- SLA calculations
- cost visibility
- cost allocation rules
- reliability thresholds
- health thresholds
- safety indicators
- leading/lagging indicators
- report templates
- report permissions
- scheduled reports
- report recipients
- export permissions
- executive dashboard layouts
- AI access permissions
- AI data scope
- AI recommendation thresholds
- forecasting periods
- alert thresholds
- tenant/venue benchmarking rules

Critical Requirement KPI Definition Governance

Softlabs should **not hardcode formulas without configuration**.

For example:

Availability

One client may calculate:

Operating Time Downtime and Operating Time

while another may exclude:

Planned Maintenance

from downtime.

Therefore TICVAI should provide governed KPI definitions.

The dashboard should show:

Availability: 98.4%

and allow authorized users to understand:

Formula

Included Downtime Types

Excluded Downtime Types

Measurement Period

This prevents disputes over KPI numbers.

Critical Requirement MTBF and MTTR

These two metrics should be native to TICVAI.

MTBF

Mean Time Between Failures

Helps answer:

How reliable is the asset?

Higher is generally better.

MTTR

Mean Time to Repair

Helps answer:

How quickly can we restore it?

Lower is generally better.

Combined:

MTBF ↓

•

MTTR ↑

=

High Reliability Concern

TICVAI should identify these patterns automatically.

Critical Requirement Maintenance Backlog

The analytics must provide a clear maintenance backlog.

Not only:

“286 Open Work Orders”

but:

Backlog by Age

0–1 day

2–3 days

4–7 days

8–30 days

30+ days

and:

Backlog by Risk**P1****P2****P3****P4****P5**

A backlog of 100 low-priority cosmetic jobs is very different from five unresolved safety-critical jobs.

Critical Requirement Asset Reliability Score

I recommend TICVAI create a configurable:

Asset Reliability Score 0 to 100

Potential inputs:

Availability**Failure Frequency****MTBF****MTTR****Repeat Failures****Maintenance Cost****PM Compliance****Safety Findings****Asset Age**

Example:

Pump P 08

Reliability Score:

62 and 100

● **At Risk**

Reasons:

- Failure frequency ↑
- MTBF ↓
- Maintenance cost ↑
- Repeat bearing failure
- Asset approaching replacement period

This gives executives a much simpler way to identify problem assets.

Critical Requirement Operational Impact

TICVAI's maintenance reporting should go beyond engineering statistics.

For destination and venue clients, management ultimately cares about:

What did the maintenance failure do to the operation?

Therefore Board 8 should connect maintenance events to:

Attraction Downtime**Admission Capacity****Ride Capacity****Retail and F and B Availability****Guest Experience**

and, where appropriate:

Revenue Impact

Example:

Gate 04 Failure

Technical Downtime:

2h 18m

Operational Impact:

Entrance capacity reduced by 20%

That is much more useful to management than downtime alone.

Critical Requirement Cross Venue Benchmarking

For multi-venue clients, TICVAI should allow authorized enterprise users to compare:

Venue A vs Venue B vs Venue C

using normalized KPIs such as:

Availability

PM Compliance

Failure Rate

MTTR

Maintenance Cost per Asset

Safety Compliance

SLA

This can reveal differences in:

- maintenance practices,
- staffing,
- vendor performance,
- asset quality,
- or operational conditions.

Tenant isolation must always be respected.

Relationship Across Boards 1 7

Board 8 consumes data from the complete maintenance architecture:

Board 1 Assets

Provides:

Asset estate lifecycle

↓

Board 2 Preventive Maintenance

Provides:

Planned maintenance compliance

↓

Board 3 Corrective Maintenance

Provides:

Failures RCA emergency events

↓

Board 4 Work Orders

Provides:

Execution labor repair times

↓

Board 5 Safety

Provides:

Incidents hazards compliance

↓

Board 6 Spare Parts

Provides:

Consumption stock risk

↓

Board 7 Resources and Vendors

Provides:

Workforce vendors procurement dependencies

↓

Board 8 Intelligence

Provides:

Analyze → Explain → Predict → Recommend

This should be the architecture shown clearly to Softlabs.

Relationship With TICVAI Reporting and AIBoard 8 should **not create a completely separate BI technology stack.**

The recommended architecture is:

Maintenance and Safety Data

↓

TICVAI Reporting and Analytics Data Layer

↓

Dashboard and KPI Engine

↓

Power BI and Embedded BI where required

↓

TICVAI AI Analytics Layer

↓

Natural Language Query Insights Forecasts

This allows the same analytics strategy to serve Ticketing, Finance, CRM, Access Control, Device Management, Maintenance and future modules.

AI Explainability

This is particularly important.

If AI says:

“Pump P-08 is at high risk of failure.”

the user should be able to see:

Why?**4 failures in 30 days****MTBF decreased 38****Bearing temperature trend increased****Repeat fault detected****Asset age: 85% of planned life****Source Records****FLT-00182****WO-002184****WO-002014**

INS-00428

AI insights should therefore be **drillable back to the operational evidence**.

AI Governance

AI can:

- Summarize performance
- identify trends
- detect anomalies
- predict risks
- forecast demand
- identify likely root causes
- recommend maintenance frequency changes
- recommend stock changes
- identify replacement candidates
- identify vendor/resource bottlenecks

AI should **not autonomously**:

- close work orders,
- declare assets safe,
- retire assets,
- change safety compliance,
- approve procurement,
- change approved PM frequencies,
- approve return to service,
- or make financial commitments.

These remain governed human decisions.

RBAC and Data Governance

Permissions should distinguish:

View Operational Analytics

View Executive Analytics

View Safety Analytics

View Financial Maintenance Cost

View Workforce Performance

View Vendor Performance

Build Reports

Schedule Reports

Export Data

Use AI Query

View AI Recommendations

View Cross Venue Benchmarking

Sensitive information such as:

Employee performance

Supplier pricing

Maintenance costs

Safety incidents

may require separate permissions.

All reporting must enforce:

Tenant → Venue → Role → Data Scope

Board 8 Acceptance Criteria

Board 8 is accepted when TICVAI can successfully:

Collect → Consolidate → Measure → Compare → Drill Down → Report → Forecast → Detect Risk → Explain → Recommend

across the complete Maintenance & Safety Management module.

Management must be able to move from:

Maintenance performance is getting worse

to:

Why

and then drill down:

Venue → Asset Category → Asset → Failure → Work Order → Part/Vendor/Technician → Root Cause

without leaving the governed TICVAI data model.

The complete intelligence chain should be:

Asset → Maintenance → Failure → Work Order → Safety → Parts → Resources → Cost → Downtime → SLA → KPI → AI Insight → Recommended Action → Source Evidence

Board Boundary

Board 1 — Asset identity & lifecycle

Board 2 — Preventive maintenance planning

Board 3 — Corrective & emergency maintenance

Board 4 — Work-order execution & field operations

Board 5 — Safety, inspection & compliance

Board 6 — Spare parts & maintenance inventory

Board 7 — Maintenance procurement, resources & vendor operations

Board 8 — Maintenance intelligence, analytics & reporting

Module 17 Summary

With Board 8, the full **Maintenance & Safety Management module is complete:**

8 Boards × 10 Screens = 80 Screens

covering the full lifecycle:

Asset → Prevent → Detect → Repair → Protect → Supply → Resource → Analyze.